BATO SPRING RESOURCE UTILIZATION OPTIMIZATION AND MANAGEMENT SYSTEM INTEGRATING MACHINE LEARNING ALGORITHM

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In partial fulfillment of the requirement for the degree
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ABSTRACT

The goal of this study, "Bato Spring Resource Utilization Optimization and Management System Integrating Machine Learning Algorithm," is to create an intelligent, web-based management system that will enhance Bato Spring Resort's overall operational effectiveness and resource management. The use of cutting-edge technologies that enable data-driven decision-making is required due to the growing need for efficient resource planning in resort operations. To improve forecasting accuracy and maximize resource utilization, the suggested system makes use of both historical and current operational data, such as guest reservations, occupancy rates, and facility usage records. Using patterns found in prior resort data, a supervised machine learning algorithm was used to forecast guest volume. These predictions serve as the basis for an integrated optimization algorithm that dynamically allocates critical resources such as manpower, utilities, and resort facilities in accordance with anticipated demand levels. By aligning operational planning with predictive insights, the system helps minimize resource wastage, reduce operational costs, and improve service delivery. The developed system was evaluated using the ISO/IEC 25010 software quality model, with particular emphasis on functionality, reliability, and usability. Additionally, the Technology Acceptance Model (TAM**)** was utilized to assess user satisfaction, perceived ease of use, and perceived usefulness among resort administrators and staff. Evaluation results indicate that the system achieved high accuracy in guest volume prediction and demonstrated effective optimization of resort resources, leading to a noticeable reduction in operational inefficiencies. Furthermore, TAM findings revealed a high level of user acceptance and a strong positive behavioral intention toward system adoption. Overall, the study demonstrates that integrating machine learning and optimization algorithms into resort management systems significantly enhances operational performance, supports sustainable resource utilization, improves managerial decision-making, and promotes positive user acceptance of intelligent information systems.

Keywords: *Machine Learning, Resource Optimization, Resort Management System, Guest Volume Prediction, Web-Based System, ISO/IEC 25010, Technology Acceptance Model (TAM)*

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DEFINITION OF TERMS

Artificial Intelligence (AI)	A subfield of computer science devoted to creating machines capable of performing tasks that typically require human intelligence. Used as the primary technology in this investigation that allows the resort management system to make intelligent decisions and perform predictive analytics.
Calendar Heatmap	A data visualization that displays values over days, weeks, or months through color intensity. Used to visualize guest volume patterns and seasonal trends in Bato Spring Resort visitor data.
Cottage Utilization	The measurement of how frequently cottages or lodging facilities are occupied. Used by the system to monitor availability and optimize allocation of cottages for incoming guests.
Data Preprocessing	Prior to model training, raw data must be cleaned, transformed, and organized. used to prepare guest records for machine learning algorithms, eliminate errors, and fill in missing values.
Data Visualization	The graphical presentation of data to communicate information clearly through charts or graphs. Used in the dashboard to show guest trends, predictions, and resource usage.
Decision Tree	In order to make predictions, a machine learning algorithm divides data into branches according to conditions. a model that helps forecast visitor numbers by analyzing guest data.
Guest Volume Prediction	A forecasting process that estimates the number of expected guests using historical data. One of the system's primary outputs, used to guide resource allocation at Bato Spring Resort.
Heatmap	A visualization technique that uses color gradients to represent data values across a matrix. Utilized to show patterns in guest attendance and peak/off-peak periods.

Machine Learning (ML)	This area without explicit programming, computers can learn from data and improve their performance thanks to artificial intelligence. powers the resort management system's guest forecasting feature.
Optimization Algorithm	A supervised learning technique used to predict continuous numerical values. Used to estimate future guest turnout based on historical patterns.
Regression Model	A computational method used to determine the best way to solve a problem under defined constraints. Helps the system determine the best resource allocation based on predicted demand.
Random Forest	An ensemble machine learning technique that creates predictions by constructing several decision trees and combining their results that are more accurate. utilized as one of the models examined to predict the number of visitors to Bato Spring Resort.
Resource Management	The planning, monitoring, and allocation of available assets to ensure efficiency. The system’s function that oversees cottages, rooms, tables, and staff availability at the resort.
Simple Imputer	A preprocessing tool that fills missing data values using strategies like mean or median substitution. Applied to address incomplete guest data before training machine learning models.
Supervised Learning	A kind of machine learning where labeled input–output models are trained using data. Used for training the guest prediction model, past guest counts act as labeled data.

CHAPTER 1

INTRODUCTION

Project Context

The Philippines has steadily emerged as a prominent tourism destination in Asia, experiencing remarkable growth over the past decade in both international tourist arrivals and visitor expenditures. This upward trend highlights the country's increasing appeal to global travelers drawn by its natural beauty, rich culture, and warm hospitality. The Philippine economy and labor market greatly benefit from the tourism industry, which creates jobs and supports a number of regional businesses, including retail, hospitality, and transportation. (Aquino & Porter, 2022).

12.7% of the nation's GDP in 2019 came from tourism-related direct gross value added (TDGVA), amounting to PHP 2.48 trillion (around US$51.25 million). This represented a 10.8% increase from the previous year’s PHP 2.24 trillion (approximately US$46.29 million). The tourism industry also saw a rise in employment, with 5.71 million people working in tourism-related sectors in 2019, marking a 6.5% growth from 5.36 million in 2018. This growth reflects the expanding role of tourism in the overall job market, with tourism industries contributing 13.5% of the total employment in the country. The steady growth of both economic output and employment in tourism highlights the sector's significant impact on the Philippines' economic development and labor force dynamics (Andulana et al., 2021).

Region IV-A, also known as CALABARZON, includes the province of Laguna and is considered one of the most visited areas in the Philippines. Laguna is known for its natural

beauty, rich history, and diverse culture. From stunning waterfalls to historical landmarks, Laguna has something for everyone (Maches, 2023). Its proximity to Metro Manila also makes it a convenient getaway destination for both short-term and extended visits.

Laguna is especially attractive to tourists for destinations such as Mt. Makiling, the hot spring resorts of Los Baños, and the man-made Caliraya Lake. Pagsanjan, dubbed the “Tourist Capital of Laguna,” is also well-renowned for its “Shooting the Rapids” boat ride to the majestic Pagsanjan Falls, an experience that draws both local and international travelers (Aguado, 2025). These attractions not only offer recreation but also contribute significantly to the province’s local economy through tourism-related activities.

In addition to the locations listed, there is the Bato Spring Resort. Bato Spring Resort, located in San Pablo City, Laguna, serves as the primary site for this study. The resort traces its origins to the natural emergence of spring waters, which have long been celebrated for their therapeutic and rejuvenating properties. These pristine waters, sourced directly from the surrounding mountains, flow continuously into the resort’s natural pools, maintaining cool temperatures year-round. The mineral composition of the spring water is believed to promote various health benefits, making the resort a favored destination for wellness seekers (L., 2025).

Bato Spring Resort features a variety of natural pools designed to complement the surrounding environment. These pools cater to a wide range of guests, from shallow areas suitable for children to deeper sections ideal for swimming and immersion. The landscape is surrounded by abundant greenery and scenic views, enhancing the restorative atmosphere of the site (Orange, 2023)

One of the challenges that Bato Spring Resort currently faces is its reliance on a manual system for monitoring guest entries, where staff record visitor information using paper logs. While this traditional approach has been in use for years, it has proven to be inefficient and error-prone, particularly as the resort continues to attract more guests. This manual method lacks the ability to track real-time guest volume, making it challenging for management to anticipate visitor surges during peak seasons or prepare for lulls during off-peak periods.

As a result, the resort struggles with optimizing the allocation of essential resources such as rooms, cottages, tables, and staffs. During peak periods, this often leads to overcrowding, underprepared facilities, and overworked personnel, negatively impacting guest experience. in low seasons, resources remain underutilized, leading to unnecessary operational costs and inefficiencies. This gap in data-driven decision-making underscores the need for a more intelligent, automated, and predictive management system that can enhance operational efficiency and improve guest satisfaction.

Resource utilization and management are essential components in ensuring the efficient operation of any organization, particularly in service-oriented industries such as tourism and hospitality. Resource utilization refers to the strategic calculation of how effectively an organization’s assets—including manpower, facilities, time, and other operational resources—are being deployed to meet its objectives. Effective utilization helps in identifying inefficiencies, improving resource allocation, and optimizing productivity.

The ability of resource utilization to offer insight into the general well-being and effectiveness of operations and projects makes it crucial. By analyzing how resources are distributed and consumed, resort management can avoid bottlenecks, prevent resource underperformance or overuse, and mitigate risks that may affect service delivery. For instance, recognizing times of peak demand allows for better staff scheduling, while monitoring available cottages and rooms ensures that accommodation capacity is neither underutilized nor overstretched. These insights ultimately contribute to improved guest satisfaction and operational efficiency (Danao, 2024).

Resource utilization is a critical component of broader resource management practices, including resource planning, leveling, scheduling, and smoothing. It allows decision-makers to make informed adjustments to project plans, budgets, and timelines. The intelligent forecasting of prospective customer volumes and peak periods made possible through the incorporation of machine learning into resource management guarantees that facilities and personnel are best matched with demand. This data-driven method not only improves the process of making decisions but also maximizes return on investment (ROI) by minimizing resource wastage and ensuring efficient use of available assets (Boone, 2023).

The incorporation of machine learning (ML) technologies into current management systems is a crucial component of this shift. These innovations offer new approaches to data analysis, resource management, and process optimization, particularly in resort operations where visitor numbers can fluctuate significantly throughout the year (Putri & Athoillah, 2024). Machine learning allows machines to recognize patterns, learn from past data, and forecast without explicit programming, a subfield of artificial intelligence.

Resort and hotel management relied heavily on manual processes for managing guest information, scheduling, and resource allocation. While these methods provided a personal touch, they often lacked scalability and accuracy, particularly during peak seasons or periods of unpredictable visitor flow. With the emergence of intelligent systems, resorts can now leverage historical data to anticipate guest turnout, automate data entry, and visualize usage trends through dynamic dashboards, significantly improving operational workflows (Baber & Arsalan, 2024).

Integrating ML into resort management systems contributes not only to internal efficiency but also to enhanced guest satisfaction. Accurate prediction of visitor volume allows for better allocation of staff and resources, ensuring a smooth and comfortable experience for guests. In addition, real-time visualizations enable management to make informed decisions, quickly identify issues, and respond proactively to changing circumstances (Ma, 2024).

This study aims to address the operational challenges of Bato Spring Resort by developing a web-based system integrated with machine learning algorithms. The system focuses on predicting guest volume, optimizing resource usage, and presenting insights through interactive visualizations, ultimately contributing to more efficient and sustainable resort operations.

The development of a web-based system integrated with machine learning algorithms addresses several critical operational challenges faced by Bato Spring Resort. Traditional resort management practices often rely on manual record-keeping and intuition-based decision-making, which can lead to inefficiencies in resource allocation, staffing, and service delivery.

Project Objectives

In order to increase operational efficiency, the study intends to develop a web-based resource utilization optimization system for Bato Spring Resort by incorporating machine learning algorithm. The system evaluates the model's performance and predictive accuracy while analyzing both historical and current data, forecasting guest volume, and allocating resources efficiently.

1. To manage and optimize resource utilization at Bato Spring Resort through the creation of a web-based system with an integrated online booking feature.
2. To integrate a Machine Learning within the system capable of predicting guest volume, along with an optimization algorithm to effectively allocate and manage resort resources based on forecasted demand.
3. To evaluate the developed system using the ISO 25010 software quality model and the Technology Acceptance Model (TAM).

Project Purpose

This study's goal is to develop an intelligent and data-driven management system for Bato Spring Resort to optimize its operations and enhance resource utilization. By integrating machine learning algorithms, the system aims to provide accurate guest flow predictions, streamline data collection, and deliver dynamic visual insights to support informed decision-making. The implementation of this system seeks to address challenges related to crowd management, resource planning, and operational forecasting. The following are the key beneficiaries of the project:

Resort Management. The system empowers Bato Spring Resort's management team by providing real-time and predictive data regarding guest attendance patterns. Through a user-friendly dashboard, administrators can visualize trends, plan resources accordingly, and make proactive decisions to improve service delivery and avoid overcrowding.

Staff and Operations Team. With access to updated guest data and predictions, staff can better prepare for daily operations, ensuring proper allocation of manpower, amenities, and supplies. The data input system also allows seamless recording of guest-related information, reducing manual workload and data errors.

Guests. Guests benefit indirectly through improved services, reduced wait times, and a more organized resort environment. The optimization of resource use enhances overall guest satisfaction and safety during peak seasons.

Future Researchers. This system adds to the expanding corpus of knowledge in resource management and intelligent tourism. Researchers looking to investigate comparable machine learning applications in hospitality, tourism, or environmental resource optimization may find it to be a useful resource.

Scope and Limitation of the Study

This project is designed and developed to assist in the optimization and management of guest-related operations at Bato Spring Resort by incorporating machine learning technology. The system seeks to assist decision-making by recognizing patterns in data from the past and present to predict guest turnout, streamline data entry, and visualize trends in a user-friendly dashboard.

The system is set up as a web-based platform connected to a relational database, where all records related to guest visits are securely stored and managed. The system focuses solely on enhancing resource management at Bato Spring and includes core modules such as the Login Module, Data Input Module, Resource Management Module, and Data Visualization Module.

Login. The login is created for users or authorized resort staff. Every user must log in using a registered username and password to access system features. Upon successful login, the user is redirected to the system dashboard. The login module ensures that only approved personnel from Bato Spring Resort can utilize the system for operations.

Dashboard. The Dashboard serves as the central hub of the system, displaying the predicted number of guests per month based on the machine learning model's output after training. It offers a clear and accessible overview of anticipated visitor trends, helping staff anticipate demand and make timely decisions. In addition to forecasts, the dashboard presents essential information about Bato Spring Resort, including descriptive content and images that reflect its features and ambiance. This combination of predictive insights and resort information ensures that users are well-informed and prepared for both operational planning and guest engagement.

Resource Management Utilization. The Resource Management Utilization feature enables real-time tracking of the resort’s facilities, including cottages, rooms, tables, and other amenities. By providing a clear view of which resources are available, occupied, or reserved, the system helps staff manage accommodations more efficiently. This proactive approach allows for faster turnaround times, reduces double bookings, and ensures that resources are allocated based on actual demand.

Data Input. This enables users to add and manage guest-related information, such as visitation dates, guest counts, and other pertinent operational metrics. It is essential for updating the system's database in real time, which enhances the precision of the machine learning forecasts. The interface is designed to be intuitive, allowing staff to input new records, edit existing entries, and maintain data integrity. The information submitted through this module feeds directly into the predictive model, ensuring that the system adapts to recent changes in guest behavior and resort activity.

Data Visualization. The visualization provides an interactive way to monitor and analyze trends within Bato Spring Resort. It generates graphical representations of guest visit patterns, seasonal fluctuations, and predicted versus actual attendance. These visual elements assist management when making decisions based on data regarding resource allocation, staffing, and facility preparation. The dashboard is automatically updated as new records are submitted, allowing for a dynamic and real-time view of operational trends.

Limitation of the Study

The system developed in this study is limited exclusively to Bato Spring Resort and is not intended for use in other establishments or resorts. Its features, data models, and predictive algorithms are customized based on the resort's unique operational patterns and data availability. Booking, payment, and financial transaction features are not included in the system. The consistency and quality of the data supplied by resort employees determine how accurate predictions are. Only authorized personnel should be able to use the system, and public access is restricted.

CHAPTER 2

REVIEW OF RELATED LITERATURE, STUDIES, ANDSYSTEMS

Resource Utilization and Management

Efficient resource utilization is fundamental to optimizing operational performance, maximizing profitability, and achieving organizational sustainability goals. One of the most widely used techniques for systematic resource management is Linear Programming (LP), which focuses on optimizing an objective function subject to linear constraints (Splint International Journal of Professionals, 2024). LP provides organizations with a structured decision-making framework, allowing them to allocate scarce resources effectively across competing activities. Traditional solution methods such as the Graphical Method and Simplex Method have long supported LP applications.

The Graphical Method is ideal for problems involving two variables, providing a visual understanding of feasible regions and optimal points, while the Simplex Method is capable of solving larger, more complex problems but often requires considerable time and computational effort. The advancement of computational tools such as Excel Solver has further enhanced the practicality of LP, making it accessible for solving problems of any size with improved speed and efficiency.

In addition to improving internal operations, Linear Programming also supports resource planning aligned with environmental sustainability objectives. Idisi and Ogumeyo (2024) demonstrated the application of LP in environmental management by developing a model to optimize ammonia production while minimizing both water pollution and operational costs. Their study revealed important trade-offs: stricter waste discharge.

Their LP model offered a framework for decision-makers to balance economic costs with ecological conservation efforts, illustrating the growing role of optimization techniques in guiding sustainable urban resource management.

In the construction sector, Sujithra and Gopan (2022) investigated how Linear Programming could optimize resource allocation to maximize profits and minimize project costs. Their research collected real-world data from the Public Works Department and applied LP techniques to derive optimal solutions for resource management in building construction projects. Furthermore, the LP results are compared with outcomes generated using PRIMAVERA P6, a widely recognized project management software. The comparison demonstrated that LP solutions provided equally effective, and in some cases more flexible, options for project scheduling and financial planning. The study concluded that LP is a highly adoptable and practical tool for the construction industry, offering managers a reliable method for organizing project timelines, minimizing expenses, and achieving better resource distribution.

Machine Learning

Understanding and creating methods that allow machines to "learn"—that is, methods that use data to improve computer performance on a specific set of tasks—is the focus of the field of ML, or machine learning. Computers can learn without explicit programming thanks to a branch of artificial intelligence called machine learning. To create models, machine learning algorithms employ sample data, also known as training data that, without being specifically programmed to do so, is capable of making decisions or predictions. Computers learn from provided data to carry out particular tasks in machine learning programs. It is possible to program algorithms that instruct

carry out all necessary steps in order to solve simple tasks; this eliminates the need for learning on the part of the computer (Sterne & Razlogova, 2019).

ML has experienced significant growth in recent years, exerting a substantial influence across various industries. Although frequently used interchangeably, it's crucial to remember that AI and ML are different ideas. Machine learning stands out as a rapidly advancing field and a way to achieve AI. The evolution of Machine Learning has already reshaped people's interactive experiences, and its continued progression holds the promise of further influencing our future lifestyles. Notably, Machine Learning serves as the technological foundation for various applications, including chatbots, predictive text, language translation apps, and more (Mishra, 2024).

Supervised learning

In supervised learning, "Labeled" training data is employed in training machines. Machines forecast the result based on this labeled data. The phrase "labeled data" denotes the fact that specific input data is already linked to the appropriate output. The training data acts as the supervisor in supervised learning, instructing the machines on how to accurately forecast the result. This is similar to how a student learns while being guided by a teacher. Using a model that can make predictions based on a labeled dataset, the machine learns under supervision in supervised learning. A labeled dataset is one in which the target answers are already known (Tiwari, 2022).

During the supervised learning process, both input and output data are fed into the model for machine learning. A supervised learning algorithm looks for a mapping function that links the input variable (x) to the output variable (y). Several practical uses, including spam filtering, risk assessment, image classification, and fraud detection and more, use supervised learning (Liu & Wu, 2012).

Data Preprocessing

Preprocessing data involves changing unprocessed data into an understandable format, and it is a crucial step since working with raw data is impractical. Prior to utilizing algorithms for machine learning or data mining, assessing the data's quality is crucial. Sorting and filtering raw data to remove superfluous and erroneous information is part of data preparation, also referred to as data cleaning (Akimov et al., 2022).

Raw data undergoes scrutiny for errors, duplication, miscalculations, or missing values, ultimately being transformed into a suitable format for additional processing as well as analysis. This process is essential to ensure that the processing unit receives only the best data. Data preprocessing is essential because it guarantees the consistency and quality of the training data and assess prediction models. The effectiveness of data preprocessing can significantly enhance the prediction's precision and dependability results (Roy et al., 2018).

SimpleImputer

The SimpleImputer using a range of input techniques, a dataset's missing values can be filled in using the sklearn.impute module's class. SimpleImputer can handle string-based categorical data even though it is designed to work with numerical data. A scikit-learn pipeline can incorporate SimpleImputer. The default method, "mean," substitutes the column's median value for missing values. Other options include "constant," which substitutes a constant value for missing values, and "most_frequent," which substitutes the most frequent value in the column. By supplying a list of column names, Additionally, multiple columns can be imputed at once using SimpleImputer.

SimpleImputer then fills in the blanks in each of the designated columns. It's crucial to think about whether the best is imputing choice for your data when using SimpleImputer.

Random Oversampling

Using a method known as examples from the minority groups are arbitrarily duplicated and included in the instruction dataset through random oversampling to address class imbalance. This method selects examples from the minority class at random with replacement, so some examples might be selected more than once. This can be especially useful for algorithms used in machine learning that are sensitive to unbalanced data, like support vector machines and decision trees that depend on optimal data splits, or artificial neural networks (ANNs) that employ stochastic gradient descent.

However, while random oversampling can help balance the class distribution and improve model performance, it also introduces the risk of overfitting. The model may become unduly tuned to those examples by including duplicate instances of the minority class, which might lead to inadequate generalization on unobserved data. In order to prevent overfitting while maintaining the capacity of the model to successfully generalize to fresh data, it is crucial to carefully adjust the amount of oversampling (Wongvorachan et al., 2023).

Train Test Split

A key function for dividing a the scikit-learn train_test_split () method divides the dataset into training and testing subsets. The dataset must first be separated into labels (y) and features (X) in order to use this function. Four subsets are then produced by X_train, X_test, y_train, and y_test in the train_test_split () function. The testing data is utilized to assess the model's performance, while the training data is utilized to fit the model. To ensure that the model has enough data to learn while maintaining objective evaluation from the test set, the training set is typically bigger than the test set, with an 80/20 or 70/30 split.

The test_size parameter allows for explicit control over this split, while the stratify parameter helps maintain class distribution, which is crucial for handling unbalanced datasets (Stojiljković, 2025).

By ensuring that the model is assessed on data it hasn't seen during training, this technique is crucial for avoiding overfitting and underfitting. Train_test_split () contributes to a more precise evaluation of the model's capacity for generalization by dividing the training and testing data. The stratify parameter prevents the possibility of biased performance metrics when working with imbalanced datasets by ensuring that the training and test sets' class proportions correspond the original dataset. In general, An essential tool for maintaining the integrity of model evaluation and raising predictive accuracy is train_test_split () (Luo, 2023).

Heat Map

A grid of colored squares that displays a key variable's values across two axis variables is called a heatmap. Ranges are used to separate the axis variables, Similar to a bar chart or histogram, each cell's color represents the primary variable's value within that range (Słomska-Przech et al., 2021).

Heat maps offer a powerful visual method to represent data points, plotting the variables on the x and y axes can reveal patterns and correlations by using a spectrum of colors or various tones of the same hue to symbolize various values. The colors in the map help to visually distinguish between high and low values, with darker shades typically indicating higher values of the variable being represented, while lighter colors represent lower values (Davila et al., 2023). This visualization technique makes it easier to quickly identify trends, clusters, or anomalies in complex datasets.

Time Series Calendar Heatmap

A visual aid that blends the ideas of calendars and heatmaps is called a calendar heatmap. The percentage of events for each day is displayed in colored cells in a calendar layout. It is simple to identify daily and weekly patterns as well as any anomalies in the data because the days are grouped by months and years and organized into columns by week (Garg, 2021). Typically, calendar heatmaps employ a single base color, such as blue, and utilize its varying shades, tones, and tints—from light to dark—to visually represent the frequency of events for each day (Zhao et al., 2014). This design

allows for a quick, intuitive understanding of time-based trends and variations across a calendar period. Heat maps are data visualizations that display densities or values using color.

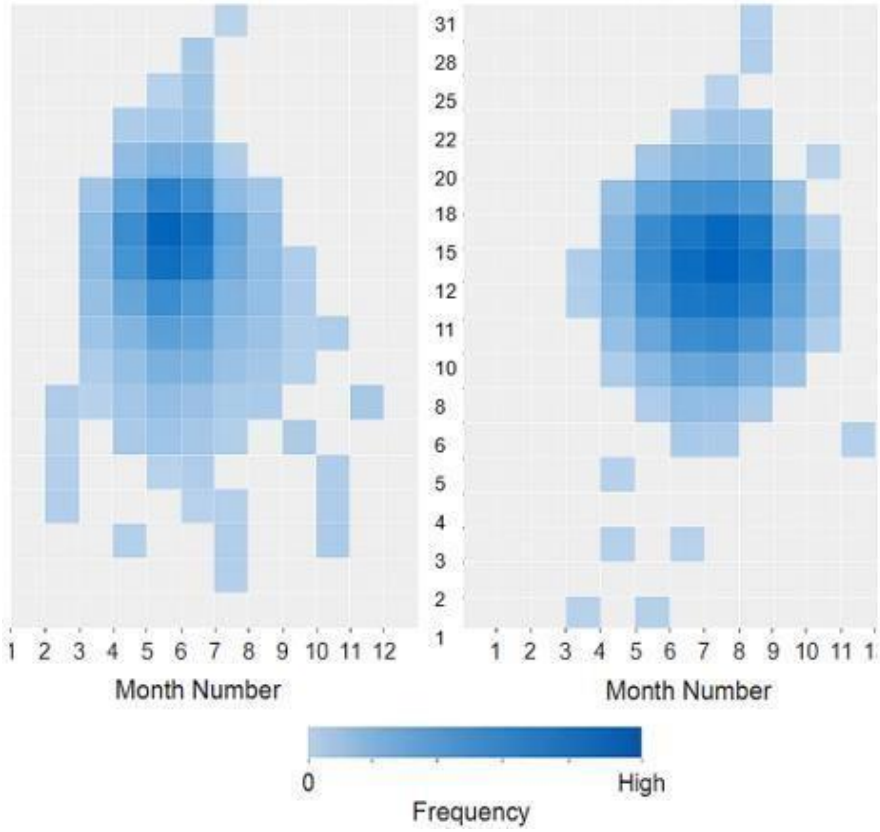


Figure 1**.** Time-Series Calendar Heatmap

Forecasting using Regression Model

Forecasting models are essential tools that leverage historical and current data to predict future outcomes. In many disciplines, such as economics, finance, and resource management, these models are essential to strategic planning and decision-making. Despite the inherent risks and uncertainties, they operate under the assumption that past relationships and patterns can be used to predict future trends (Arora, 2024). Regression analysis is one of the most popular forecasting methods, a statistical method that looks at how two or more variables relate to one another.

For example, past data on personal disposable income (X) may be used to forecast future consumer expenditures (Y). The process involves determining a "line of best fit" to estimate this relationship, which is then used to make future projections (Arora, 2024).

Time Series Forecasting is a popular quantitative technique within the larger category of forecasting models that focuses on examining data points gathered or recorded at particular time intervals—such as hourly, daily, monthly, or annual—in order to find patterns, trends, and seasonal effects. This approach is perfect for forecasting recurrent events or behaviors over time because it makes the assumption that past data patterns will persist into the future (Dastorani et al., 2016).

In industries like finance, economics, and operations, time series Regression models, exponential smoothing, and moving averages are frequently employed to predict variables such as demand, sales, or resource utilization. By examining the sequential nature of data, Time Series Forecasting provides valuable insights that help organizations anticipate future changes and make informed decisions based on observed temporal trends (Athiyarath et al., 2020).

Regression Model

Regression is a supervised learning technique in machine learning that predicts a continuous numerical value using a single or multiple independent variables. Regression analysis's objective is to ascertain a connection between the dependent (target) and independent (feature) variables, allowing the model to generate accurate predictions. Regression's target variable is continuous, meaning it can have any real value within a given range. Temperature, income, and home value are a few examples of continuous variables (Guerard, 2012).

The dependent variable (or target), which is the value to be predicted, and the independent variables (or features), which are the input variables that affect the target's value, are the two types of variables used in regression models. When forecasting home prices, for instance, the price would be the dependent variable and the number of rooms, location, and square footage could be the independent variables. Predicting a person's weight, stock price, or yearly salary are examples of situations where regression problems are frequently encountered and the output variable is a real or continuous value (Jia, 2025).

Depending on the` connections between the complexity and the variables of the data, different regression models can be applied. Assuming the linear relationship exists between the variables that are independent and dependent; this is known given that the most common method is linear regression fundamental and popular regression model. Despite its simplicity, linear regression is frequently used as a benchmark for regression tasks and forms the basis for more intricate models (Sun et al., 2023).

Decision Trees

Decision trees are visual aids for forecasting or decision-making according to a set of circumstances. The data was divided into branches, each of which symbolizes a choice or result. The decision tree starts with just one node, known as the root node, which stands for the complete dataset. Every internal node in the tree denotes. The final forecasts or results are represented by the tree's leaves (Reina- Jiménez et al., 2023).

Regression and classification problems are frequently solved with decision trees. The tree models use target variables with discrete values to classify or categorize an object in classification problems. Predictive modeling-based decision trees are popular in data mining and machine learning. By considering observations of the sample population, the model correctly determines the target value of the sample (represented by leaves) (illustrated via branches) (Mienye & Jere, 2024).

In the study conducted by (Yan-Yan et al., 2015), decision trees are described as a robust and widely adopted data mining technique, especially suitable for classification and prediction tasks. The methodology operates by dividing a dataset into smaller, more manageable subsets based on significant covariates, eventually creating an inverted tree structure with internal decision nodes, leaf nodes, and a root node. This structure

simplifies complex variable relationships, making it easier for researchers and decision- makers to interpret and apply the findings. One of the key strengths of the Decision Tree approach highlighted in this study is its non-parametric nature. Because DTs do not impose any distributional constraints, they can effectively handle large and complex datasets, in contrast to traditional statistical models that require assumptions about data distribution.

Moreover, they are capable of handling missing values and heavily skewed data without the need for preprocessing steps like imputation or data transformation. DTs are also noted for their robustness to outliers, and their intuitive visual structure enhances interpretability, making them accessible even to those without extensive statistical training. The study also introduces several well-known DT algorithms such as CART, C4.5, CHAID, and QUEST, and points out the availability of software tools like SPSS and SAS that can be used to visualize decision trees effectively. In the comprehensive review conducted by Abdulqader and Abdulazeez (2024), Decision Tree algorithms are recognized as transformative tools in the healthcare industry, particularly in the areas of diagnostics, prognosis, and health monitoring. The study emphasizes how DTs simplify complex decision-making processes, making them ideal for examining large and diverse medical datasets, such as genetic data, Real-time patient data and electronic health records. One of the main benefits of Decision Trees highlighted in this review is their ability to improve diagnostic accuracy and forecast the results of patients with greater precision. Their structured, rule-based approach allows healthcare professionals to interpret model outputs with ease, facilitating more informed clinical decisions.

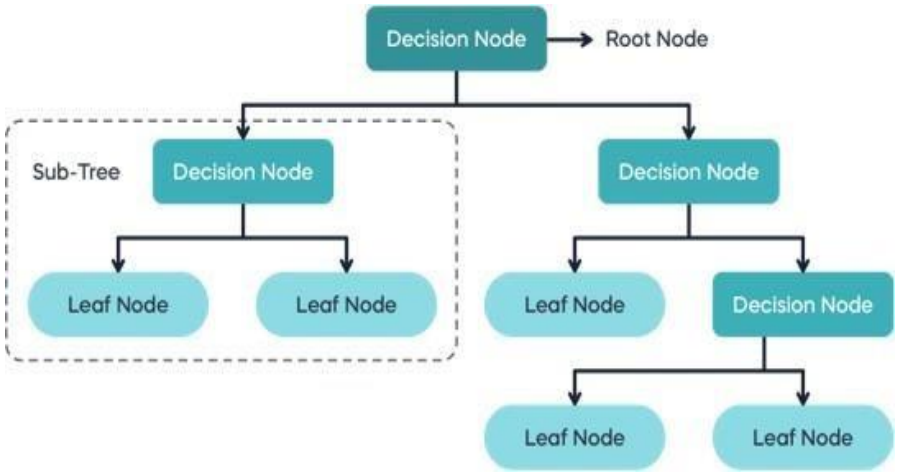


Figure 2. Decision trees model

Random Forest

An algorithm for supervised learning is called Random Forest. An ensemble the "forest" it creates is made up of decision trees, usually trained using the bagging method. The fundamental idea behind the bagging approach is that combining different learning models improve the outcome. Regression and classification issues can both be resolved with Random Forest. Its foundation is ensemble learning, which boosts the model's effectiveness by combining several classifiers to address a challenging issue (Rakhra et al., 2021). The well-known machine learning algorithm, Leo Breiman and Adele Cutler created Random Forest. It produces just one outcome by combining the output of multiple decision trees. Its popularity has been fueled by its adaptability and simplicity of use, and effectiveness as a random forest classifier because it is capable of handling both regression and classification issues. This article explains how the random forest algorithm operates, its benefits, random forest regression techniques, how it varies from other algorithms, and how to apply it. The random forest algorithm is an extension of the bagging technique that generates a combining feature randomness and bagging to create an uncorrelated forest of decision trees. The random forest algorithm has been applied to enhance business decision-making in a number of industries. Even without hyperparameter tuning, versatile and user-friendly machine learning algorithm, Random Forest typically yields outstanding outcomes. Due to its adaptability and simplicity of use (It can be used for tasks involving both regression and classification), it is also one of the most popular algorithms.

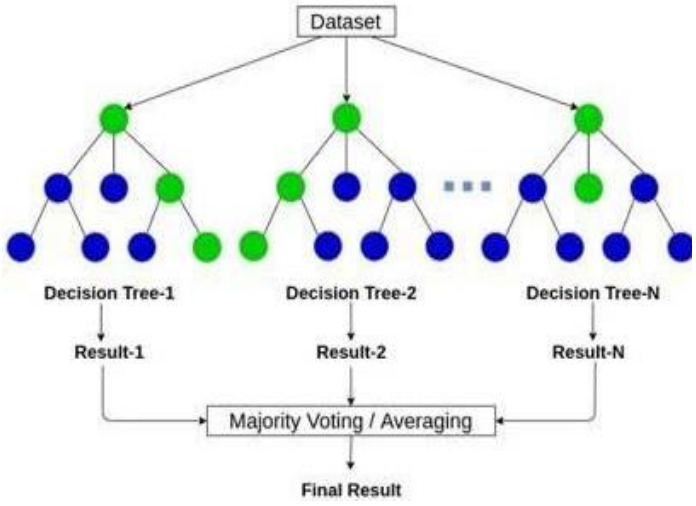


Figure 3. Sample of Random Forest

To put it briefly, Random Forest is a classifier that uses several decision trees on various subsets of the dataset and averages them to increase the dataset's expected accuracy. Instead of relying on solely on one decision tree, the random forest collects the results from all of them and uses the majority votes of forecasts to predict the outcome (Moore et al., 2019).

In the study conducted by El et al. (2021), the performance of Decision Tree (DT), Random Forest (RF), Gradient Boosted Trees (GBT), and the conventional ARIMA (0,1,1) model was compared by the researchers as they investigated the use of tree-based techniques for monthly gold price forecasting. The Root Mean Squared Error (RMSE) metric was used to assess forecasting accuracy. RF outperformed DT, GBT, and ARIMA among the tested models, with the lowest RMSE of 38.52. The authors concluded that RF is a more efficient and accurate method for forecasting monthly gold prices compared to the other models. They also proposed future work involving hybrid methods that integrate ARIMA with tree-based techniques to leverage the strengths of both approaches.

In the study conducted by Naing and Htike (2015), a Random Forest model was employed to forecast monthly temperature variations in Kuala Lumpur using historical time series data from the years 2000 to 2012. The model, which relies on the aggregation of multiple regression trees (CART), effectively addressed challenges such as no stationarity, heteroscedasticity, and seasonal trends in the data. A number of other forecasting methods, such as K-Nearest Neighbors, Least Median Square Regression, Multilayer Perceptron (MLP) neural networks and Radial Basis Function (RBF) networks, are used to compare the Random Forest performance of the model. The assessment showed that the Random Forest model achieved high forecasting accuracy using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). The findings suggest that the model provides reliable and efficient forecasts and can be considered a valuable tool for predicting monthly temperature variations.

XGBoost

Decision trees with gradient boosting are used by the supervised XGBoost (eXtreme Gradient Boosting) learning boosting algorithm that makes use of gradient descent, a distributed, open-source machine learning library. Through the use of an internal algorithm that aggregates the output from several distinct trees, can generate accurate predictions (M'hamdi et al., 2024). The ranking of the input features is simultaneously displayed by the model. Furthermore, XGBoost outperforms other classifiers in terms of minimizing running time through distributed and parallel computation, preventing overfitting, and efficiently handling missing values (Cortez et al., 2023).

where nn is the quantity of training data, at the ith instance, the feature vector and its label are denoted by xi and yi., $yk-1$ is the ith instance prediction at the t−1th iteration, fk is a new tree that uses xi to classify the ith instance and represents the

t−1th iteration, fk is a new tree that uses xi to classify the ith instance and represents the regularization term that penalizes the complexity of the new tree, and l is a loss function that computes the new tree output plus the discrepancy between the label and the final forecast.

Andrabi et al. (2022) looked into several machine learning algorithms for sales forecasting to help retailers forecast market demand in the future and maximize stock levels. The Walmart Sales dataset from Kaggle, which has more than 400,000 records and about 20 features, was used by the researchers. Several algorithms are used after feature engineering and data cleaning, including Random Forest Regressor (both with and without adjusted hyperparameters), Ridge Regression, Decision Tree Regressor, and Linear Regression, and eXtreme Gradient Boosting (XGBoost). Among these, XGBoost with adjusted hyperparameters performed better than the other models, exhibiting both high predictive accuracy and low processing and memory consumption.

In the study conducted by Bentéjac et. al (2019), an empirical analysis comparing XGBoost with Random Forest and traditional Gradient Boosting across various classification tasks. While Gradient Boosting achieved the highest accuracy overall, the performance differences between the models using default settings are not statistically significant.

However, XGBoost and Gradient Boosting performed poorly without parameter tuning, highlighting the importance of careful hyperparameter optimization. XGBoost's flexibility and scalability are emphasized, though its tuning process was found to be computationally intensive—contributing to over 99.9% of training time. Despite this, the tuning burden can be eased with a reduced parameter grid. The study concluded that XGBoost is a powerful method, but not universally superior, and excels particularly when finely tuned, as often seen in competitive machine learning scenarios.

Optimization Algorithm

They enable models to learn from data by changing internal parameters to lower error or improve predictive accuracy, optimization algorithms are crucial for both machine learning and deep learning (Ahmad et al., 2023). These algorithms are crucial for resolving challenging real-world issues because they are mathematical methods for determining the optimal solution within a specified set of limitations or goals. Their main goal is to maximize an objective function or minimize a defined loss function in order to iteratively improve model performance and guarantee that the adapts well to fresh, unobserved data. Without optimization algorithms, the training process of machine learning and deep learning models would be ineffective and unstable, ultimately leading to poor predictive results (Zaheer & Shaziya, 2019). As highlighted in various studies, optimization is fundamental not only for developing accurate predictive models but also for addressing broader engineering and operational challenges where choosing the best solution from many alternatives is required (Abdulrahman et al., 2024).

Optimization methodologies offer structured and intelligent strategies that are widely applicable across disciplines.

They allow researchers and practitioners to handle resource constraints, improve decision-making, and systematically evaluate trade-offs within complex systems (Mohammadi & Sheikholeslam, 2023).

As real-world challenges continue to evolve in complexity, the role of optimization algorithms has expanded beyond academic interest to critical applications in fields like engineering, business, and artificial intelligence. These algorithms are designed to identify the most effective solutions by systematically analyzing possible outcomes based on precisely defined objectives and constraints (Joksimovic et al., 2023).

Linear Programming Problem

Linear Programming Problem (LPP) is a widely recognized mathematical approach used for optimizing a linear objective function while adhering to a set of linear constraints, which can be either equalities or inequalities. As noted by Kunwar and Sapkota (2022), optimization lies at the core of effective business operations, and LPP serves as one of the most fundamental yet powerful tools in operations research. It provides decision makers with a systematic framework to determine the most efficient allocation of limited resources within a given set of constraints. Optimization in this context generally falls into two categories: maximization—where the aim is to achieve the highest possible output, such as maximizing profit or resource utilization—and minimization, which seeks to reduce undesirable outcomes like cost, time, or waste.

American economist Wassily Leontief and Soviet mathematician Leonid Kantorovich separately investigated the real-world uses of linear programming in the late 1930s. Leontief investigated economic applications, while Kantorovich concentrated on manufacturing schedules. For many years, their revolutionary work was mostly ignored.

According to Hsiao et al. (2021), linear programming works by optimizing an objective function formed from decision variables, all governed by linear equations or inequalities. For a problem to qualify as an LPP, it must satisfy several key assumptions: linearity of relationships, additivity of contributions, divisibility of decision variables (allowing them to take fractional values), and constancy of coefficients over the planning horizon. When these conditions are met, LPP can be applied using various solution methods. The graphical method is suitable for problems involving only two variables, while more complex scenarios with multiple variables are typically solved using the Simplex Method or digital tools such as Excel Solver.

Synthesis and Relevance of the Study

This study introduces an intelligent, machine learning–driven web-based management system designed to optimize operations and resource allocation at Bato Spring Resort. Through predictive analytics, real-time data input, and dynamic visualizations, the system offers a data-centric approach to improving operational efficiency, forecasting guest volume, and enhancing overall service delivery. By integrating machine learning models, the system can recognize historical patterns and generate accurate predictions of visitor turnout, thus supporting proactive decision-making and efficient resource planning. This study's main contribution is its illustration of the practical applications of contemporary technology, particularly artificial intelligence and machine learning, in the hospitality and tourism sector. The study highlights the increasing demand for intelligent systems that do more than just automate procedures but also offer insights that result in operations that are more guest-focused and sustainable.

It addresses the operational challenges unique to resorts, particularly the fluctuations in guest volume and the complexity of managing resources efficiently during peak seasons.

The relevance of this study is underscored by its potential to transform traditional resort management practices. At a time when the tourism industry is being reshaped by technological advancements and heightened customer expectations, the proposed system serves as a timely and practical solution. It empowers resort management with tools for better planning, minimizes manual errors, reduces workload, and ultimately contributes to a more satisfying guest experience.

The system offers useful insights for both industry practitioners and academic researchers, establishing the foundation for future studies in intelligent tourism and digital resource management. This study highlights the significance of embracing digital transformation in managing contemporary hospitality establishments like Bato Spring Resort by addressing both the operational and technological gaps.

Additionally, by showcasing the usefulness and efficiency of machine learning-driven systems in small- to medium-sized resort operations, this study advances the expanding fields of intelligent tourism and smart resource management. This study demonstrates how comparable data-driven and predictive technologies can be successfully adapted to smaller operational contexts with limited resources, whereas many previous studies and implementations of intelligent systems are focused on large hotels, urban tourism hubs, or multinational hospitality chains. By doing this, the study fills a significant vacuum in the literature and offers empirical proof that machine learning approaches are both practical and advantageous for resort environments that are modestly scaled.

Future researchers, system developers, and hospitality professionals who want to implement or improve intelligent decision-support systems in the travel and tourism sector will find great value in the study's conclusions. In particular, the study shows how automated data processing, optimization strategies, and predictive analytics can enhance workforce planning, resource allocation, and service scheduling. These enhancements lead to higher operational effectiveness, lower operating expenses, and better use of available resources.

Literature Matrix

Table 1. RRL Matrix

AUTHOR & YEAR	TITLE	METHODS/ALGORITHMS, ARCHITECTURES USED	FINDINGS
Abdulqader, H.A. & Abdulazeez, A.M. (2024)	A Review on Decision Tree Algorithm in Healthcare Applications	Decision Tree Algorithms, Integration with Machine Learning Techniques	The study highlights how decision trees simplify complex medical decision-making, improve diagnostic accuracy, and support prognosis and health monitoring. Decision trees effectively process diverse data types like genetic markers and electronic health records. Their integration with advanced machine learning methods enhances their analytical power, aiding in personalized treatment planning and efficient resource allocation. Challenges such as data privacy and annotation needs are acknowledged.
Smith et al., 2022	Machine Learning in Water Resource Management	Case studies, ML models like Random Forest, SVM	ML improves forecasting accuracy and resource allocation
Lopez & Chan, 2021	Smart Tourism and Natural Resource Sustainability	Surveys, IoT integration, Predictive Analysis	Smart systems enhance resource sustainability in tourist areas
Wang et al., 2020	Optimization of Natural Spring Water Systems Using AI	Simulation models, Neural Networks	Neural networks can predict resource consumption patterns
Madaan, M., Kumar, A., Keshri, C., & Jain, R. (2021)	Loan Default Prediction Using Decision Trees and Random Forest: A Comparative Study	Random Forest and Decision Tree	The Random Forest classifier achieved 80% accuracy, outperforming the Decision Tree's 73%. The study demonstrated. Random Forest's greater reliability for this classification problem. However, the imbalance in the dataset (few defaulters) was noted as a limitation, and expanding the dataset for future work was suggested to improve model accuracy.

Hafsa Taya, E. (2022)	Forecasting Regional Tourism Demand in Morocco from Traditional and AI-Based Methods to Ensemble Modeling	ARIMA, AR, Linear Regression, SVR, XGBoost, LSTM, Ensemble Learning	The study forecasts the number of tourist arrivals to the Marrakech-Safi region using both traditional methods (ARIMA, AR, linear regression) and AI-based techniques (SVR, XGBoost, LSTM). Hybrid models combining conventional and AI-based techniques using ensemble learning outperformed both individual conventional and AI- based methods, showing that hybrid approaches can overcome the limitations of each method.
Zhang, B. et al., (2020)	A deep learning approach for daily tourist flow forecasting with consumer search data	Long Short-Term Memory (LSTM)	The study introduces the LSTM network for handling complex time series forecasting in tourism. Using daily tourist flow and consumer search data from Jiuzhaigou (China) from 8 October 2013 to 7 August 2017, the method was empirically tested. The results show that LSTM outperforms other models, with the best tatistical performance in both training and test sets.
Shabri, A. et al., (2020)	Combining Deep Neural Network and Fourier Series for Tourist Arrivals Forecasting	Deep Neural Network (DNN), Fourier Series	The study presents a hybrid model combining DNN with the Fourier series method for forecasting tourist arrivals. The DNN is used to generate forecasts, and Fourier series is applied to fit the residual error.
(2022)	Learning Models for Long-Term Tourism Demand Forecasting	Memory (LSTM), with and without integration of exogenous variables	models for long-term forecasting of hotel bookings, tested on real data from three hotels in Greece. One model used only historical booking data, while the other integrated weather data.

Han, S., Williamson, B. D., & Fong, Y. (2021)	Improving Random Forest Predictions in Small Datasets from Two-Phase Sampling Designs	Used Random Forest (RF), Generalized Linear Models (GLM), and stacking techniques. Explored effects of variable screening, class balancing, inverse probability weighting (IPW), and hyperparameter tuning on RF performance.	Variable screening improved RF performance significantly. Class balancing improved performance only when variable screening was not applied; otherwise, it worsened results. IPW was beneficial when combined with screening. Hyperparameter tuning was ineffective in small datasets. RF underperformed compared to GLMs for some feature sets, but stacking RF and GLM led to improved results.
Houria, L. et al., (2020)	Deep Learning Framework for Forecasting Tourism Demand	Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU); compared with Support Vector Regression (SVR) and Artificial Neural Networks (ANN)	Demonstrated that both LSTM and GRU outperform traditional models (SVR and ANN) in forecasting accuracy for tourism demand. These deep learning methods effectively captured the non- linear patterns in tourist arrival data.
Zhao, S. (2023)	Customer Churn Prediction Based on the Decision Tree and Random Forest Model	Decision Tree Classifier, Random Forest Classifier	The study predicts customer churn using decision tree and random forest models. The random forest outperforms the decision tree with an accuracy of 91%. Feature selection and visualization of feature importance are used. The models are tested using an 80–20 train–test split
Shaik, N.B., Jongkittinarukorn, K., & Bingi, K. (2024)	XGBoost based enhanced predictive model for handling missinginput parameters: A case study on gas turbine	XGBoost (Extreme Gradient Boosting); evaluated using RMSE, MSE, MAE, R^2; Integrated feature importance analysis	Developed a robust XGBoost regression model to predict Energy Yield (EY) of gas turbines. Demonstrated strong performance in predicting outcomes under varied conditions and handling missing input data (e.g., CDP and OT). Feature importance analysis offered insight into key influencing factors.Model proved effective in real-time applications and outperformed other ML models in energy
Zhang et al. (2020)	Prediction of Spring Water Quality Using Machine Learning	Random Forest, SVM, ANN	Random Forest outperformed others in predicting water quality based on seasonal and land-use data.

Abdulqader & Abdulazeez (2024)	A Review on Decision Tree Algorithm in Healthcare Applications	Decision Tree algorithms; integration with Machine Learning techniques	Decision tree algorithms simplify complex decision-making and effectively process diverse datasets. Their integration with other ML methods enhances prediction accuracy, supporting optimized resource allocation despite challenges in data privacy and annotation.
Li et al. (2022)	Smart Water Resource Management Using Machine Learning	Random Forest, Support Vector Machine (SVM), IoT-based system architecture	Machine learning models significantly improved water demand prediction and resource allocation efficiency. IoT integration enabled real-time monitoring and proactive management of water resources.
Santos & Reyes (2021)	Water Resource Optimization Using Predictive Analytics	Linear Regression, Neural Networks, Cloud-based architecture	Predictive analytics improved water utilization planning and reduced wastage. Neural networks outperformed traditional statistical models in forecasting water availability.
Kumar & Patel (2020)	(2020) Intelligent Monitoring Systems for Natural Water Sources	Sensor-based architecture, ANN, Data Analytics	Intelligentmonitoring systems improved early detection of resource depletion and supported sustainable management of natural water sources through accurate predictive modeling.

Al-Jarrah, O. Y., Yoo, P. D., Muhaidat, S., Karagiannidis, G. K., & Taha, K. (2020)	Efficient Machine Learning for Big Data: A Review	Machine Learning Algorithms (Random Forest, Decision Tree, Neural Networks); Big Data Analytics Frameworks	The study reviews efficient machine learning techniques applied to large-scale datasets for resource optimization and management. Findings show that decision tree–based and ensemble models provide high interpretability and accuracy for predicting resource demand and system behavior
Rahman et al. (2022)	Machine Learning-Based Water Demand Forecasting for Sustainable Resource Management	Random Forest, Support Vector Regression (SVR), IoT-enabled monitoring architecture	The study demonstrated that machine learning models, particularly Random Forest, achieved high accuracy in forecasting water demand. The integration of IoT sensors enabled real-time data collection, which improved decision-making for sustainable water resource utilization and reduced water wastage through predictive planning.
Chen, X., Wang, Y., & Li, Z. (2021)	Machine Learning-Based Water Resource Management and Optimization Systems	Machine Learning Algorithms (Decision Trees, Random Forest, Neural Networks); Data-Driven Optimization Models	The study demonstrates that machine learning-based systems significantly improve water resource utilization efficiency through accurate demand prediction and usage pattern analysis. The models support optimized allocation, real-time monitoring, and sustainable management of water resources.

Conceptual Framework

The conceptual framework illustrates the process flow and interaction of key components involved in the development of the Bato Spring Resource Utilization Optimization and Management System Integrating Machine Learning Algorithm**.** It presents how data, machine learning, and optimization techniques are integrated to support intelligent decision-making and efficient resort operations.

The framework begins with the identification of resort resources, which include cottages, rooms, tables, and staff. Each resource is defined along with its corresponding capacity and limitations. These resources represent the operational inputs that need to be managed efficiently to meet guest demand.

The data input module of the system then collects visitor data, both past and present. This data includes resource usage logs, visitor numbers, and visit dates. Before the machine learning model uses the data, it undergoes data preprocessing, which includes cleaning, addressing missing values and getting the data ready for analysis. To guarantee precise and trustworthy model development, the processed dataset is then separated into sets for training, validation, and testing.

The machine learning component uses supervised learning techniques, specifically regression-based models, to analyze patterns from historical guest data and predict future guest volume. The trained model generates forecasts that estimate the expected number of visitors for specific time periods. These predictions serve as the primary basis for planning and decision-making within the system.

Following the prediction phase, an optimization algorithm, particularly Linear Programming, is applied. This component uses the predicted guest volume as input to determine the optimal allocation of resort resources.

The algorithm considers constraints such as resource capacity and availability, ensuring that manpower, cottages, and amenities are efficiently distributed based on anticipated demand.

The final phase of the framework is the system integration and output layer**,** where prediction results and optimized resource allocations are presented through a web-based dashboard. The dashboard provides data visualizations, real-time monitoring of resource utilization, and actionable insights for resort administrators and staff. These outputs support proactive planning, reduce manual workload, and enhance operational efficiency.

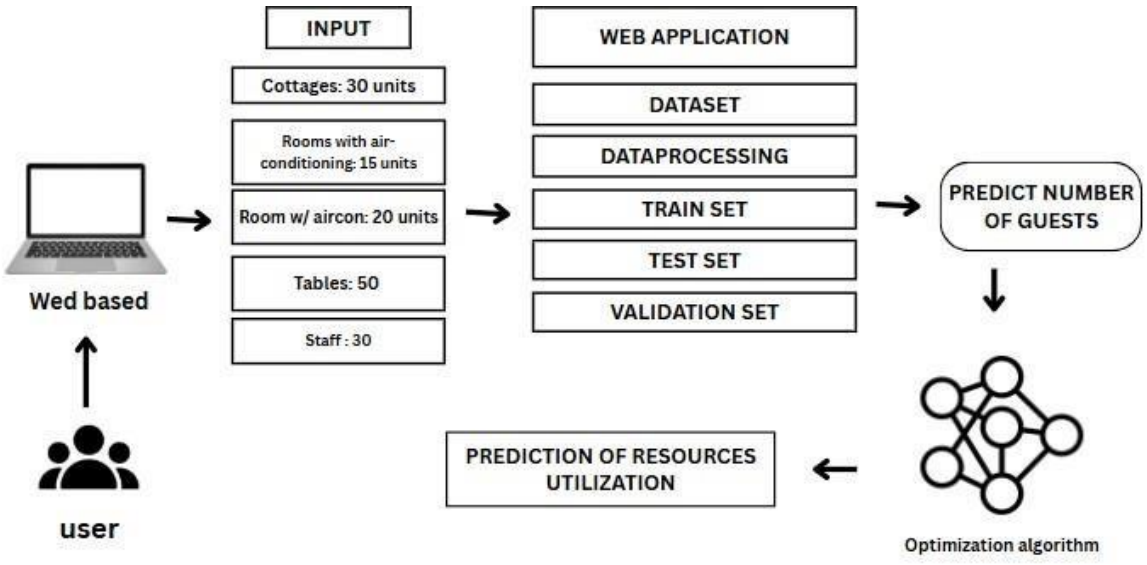


Figure 4. Conceptual Framework

CHAPTER 3

RESEARCH METHODOLOGY

Project Designs

The process begins with identifying and analyzing the resort's available resources, such as cottages, rooms, tables, and staff, while also determining the guest capacity for each based on their sizes and accommodation limits. Users can input and manage guest- related data, including the number of guests and visit dates. This data undergoes preprocessing and is split to prepare it for training a regression model. Once trained, to assess the accuracy of the model and dependability in forecasting guest volume, it is tested on unseen data. With accurate predictions in place, an optimization algorithm is applied to enhance resource utilization. The next step involves integrating the model into a software system, followed by the development of an intuitive user interface, system testing, and evaluation. After successful validation, the system is finalized and ready for deployment.

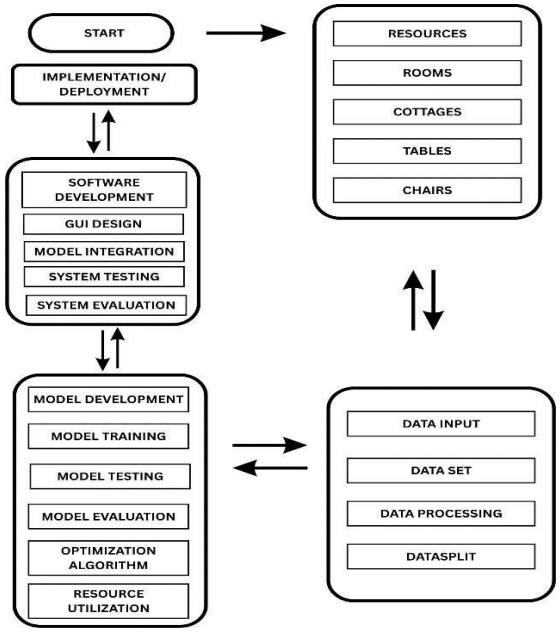


Figure 5. System Architecture

Project Locale

Figure 6 shows Bato Spring, located in Barangay San Cristobal**,** near the foothills of Mount Banahaw in San Pablo City, Laguna. Known for its clear, cold spring water**,** Bato Spring has long been a source of fresh water for local communities and tourism. This project seeks to harness the natural spring sustainably for broader water supply or eco- tourism enhancement. Bato Spring Resort serves as a popular recreational destination for both local and visiting tourists, particularly during weekends, holidays, and peak seasons. The resort offers various amenities such as cottages, rooms, tables, and other facilities designed to accommodate different group sizes and visitor needs. Due to its growing number of guests, the resort experiences fluctuating visitor volumes, which directly affect resource utilization and operational efficiency.



Figure 6. Location of Bato Spring

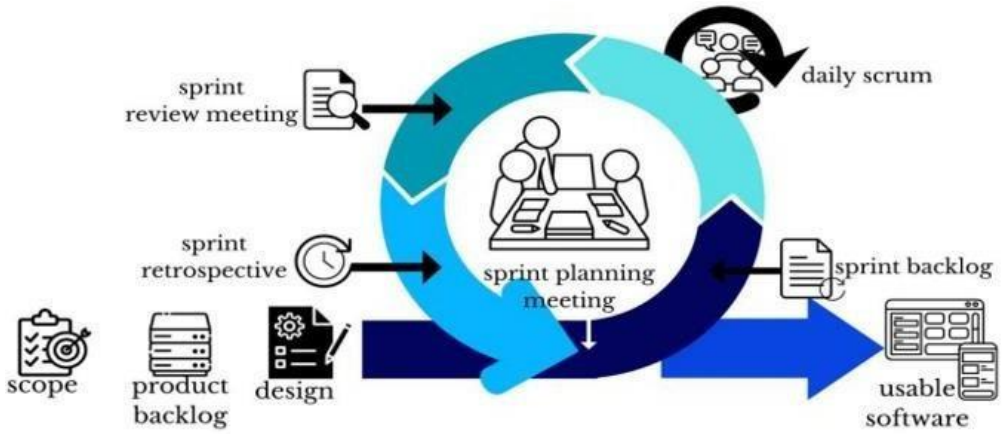
Project Development

The web-based system was designed and created utilizing the agile software development life cycle approach. The project's development is divided into several phases by the agile scrum methodology. It entails ongoing cooperation with end users and ongoing development at every level.

Agile Scrum Software Development

The development of the system for Bato Spring Resort followed the Agile Scrum Software Development methodology. This approach supported an iterative, adaptive, and collaborative process that allowed the development team to respond to feedback and evolving requirements throughout the project life cycle. The following phases and practices are implemented:

Figure 7. Agile Scrum Methodology



Scope. The initial scope of the project that concentrated on creating a web-based system designed to enhance operational efficiency at Bato Spring Resort through predictive analytics, real-time data input, and interactive visualizations. The system was built to assist resort staff and management in managing guest turnout, allocating resources effectively, and reducing manual workload.

Core modules include a secure login system for authorized personnel, a centralized dashboard that displays guest predictions and resort information, a resource management module for monitoring cottages and amenities, and a data input module that supports the continuous training of the machine learning model. The platform aimed to provide timely insights for planning and decision-making while offering a better resort experience for guests.

Design. A user-centered design approach was adopted to ensure ease of use and operational efficiency. Storyboards and interface mockups are created early in the development process to visualize the layout, navigation flow, and functionality of each module.These visual guides served as reference points throughout the system's development, helping ensure that the user experience was intuitive and aligned with the resort’s operational needs. Regular feedback loops and team discussions enabled the iterative refinement of each feature, ensuring that both functionality and design are tailored to the end users.

Product Backlog

Table 2 presents the Product Backlog, which outlines the list of essential tasks identified for the development of the resort management system. These tasks are formulated based on the features that the developers intended to include, focusing on the needs and operations of Bato Spring Resort’s administrators and staff. Each user story reflects a general yet vital functionality related to data input, guest prediction, resource monitoring, or real-time visualization. The estimated hours provide a guide for the time required to complete each task, helping to prioritize the development process. As an initial phase under the Agile Scrum methodology, the product backlog served as the foundation for planning and implementing the system’s core features through iterative development.

Table 2. Product Backlog

Priority	User Story	Est. hr/day
1	As a user, I want to securely log in to the system to access features.	2 hrs
2	As a user, I want to input guest data (visit date, number of guests, etc.).	4 hrs
3	As a user, I want to view monthly guest predictions on a dashboard.	1 hr
4	As a user, I want to monitor real-time availability of cottages and amenities.	3 hrs
5	As a user, I want to see dynamic graphs showing guest trends and seasonal patterns.	2 hrs

Sprint Planning

Table 3 presents the current Sprint Backlog, which highlights the prioritized tasks to be completed within the sprint cycle. This includes a detailed list of what needs to be accomplished before the next sprint begins. The user stories are aligned with the actual requirements and expectations of the system's end-users—primarily the resort staff and administrators. During the Daily Scrum, the developers conduct 15-minute meetings to monitor sprint progress and assess if any adjustments in the backlog or sprint goals are necessary. These meetings also help identify and resolve any roadblocks or challenges faced by the team to maintain development momentum. In the Sprint Execution phase, developers actively collaborate with the client to determine feasible deliverables and the approach to achieve the sprint objectives efficiently. This close coordination ensures that development remains aligned with the client’s expectations and allows for quick feedback and iteration.

A Sprint Review Meeting is held weekly to demonstrate the accomplishments of the team, allowing the client to provide feedback or request minor modifications or additions to ensure the system meets all intended requirements.

Within the scrum framework, sprint planning is a time-limited activity that initiates the next sprint for agile teams. Sprint planning determines what can be done in the sprint and how.

In scrum, the sprint is a set period of time during which all of the work is finished. However, before you can begin running, you have to set up the sprint The starting point, the time box's duration, and the sprint goal must all be selected. The agenda and focus are established at the sprint planning meeting, which initiates the sprint. When done properly, it also fosters an atmosphere where the group is capable of success, motivated, and challenged. The team may fail as a result of unrealistic expectations created by subpar sprint plans.

Table 3. Sprint Planning

Priority	User Story	Task Description	Est. Hours
1	As a user, I want to securely log in to the system to access features.	Design and develop the login module with authentication and security validation.	2 hrs
2	As a user, I want to input guest data (visit date, number of guests, etc.).	Design and develop guest data input form and link to database.	4 hrs
3	As a user, I want to view monthly guest predictions on a dashboard.	Integrate ML model and create a dashboard display for predictions.	1 hr
4	As a user, I want to monitor real-time availability of cottages and amenities.	Develop a real-time monitoring panel with database sync.	3 hrs
5	As a user, I want to see dynamic graphs showing guest trends and seasonal patterns.	Design and display interactive charts using guest data analytics.	2 hrs

Flowchart

The process starts with Login, where an authorized user accesses the system, and then proceeds to the Dashboard. The dashboard displays important summary information such as the number of clients per month and general details about Bato Spring Resort. From the dashboard, the user can navigate to several main modules. Information Management allows the user to add, edit, delete, and search records, with all changes stored in the database. Resources Management enables viewing available rooms and cottages, helping manage resort accommodations efficiently. Data Visualization presents information in the format of charts and bar graphs for easier analysis and decision-making. Finally, the Logout option ends the user session, leading to the End of the system process.

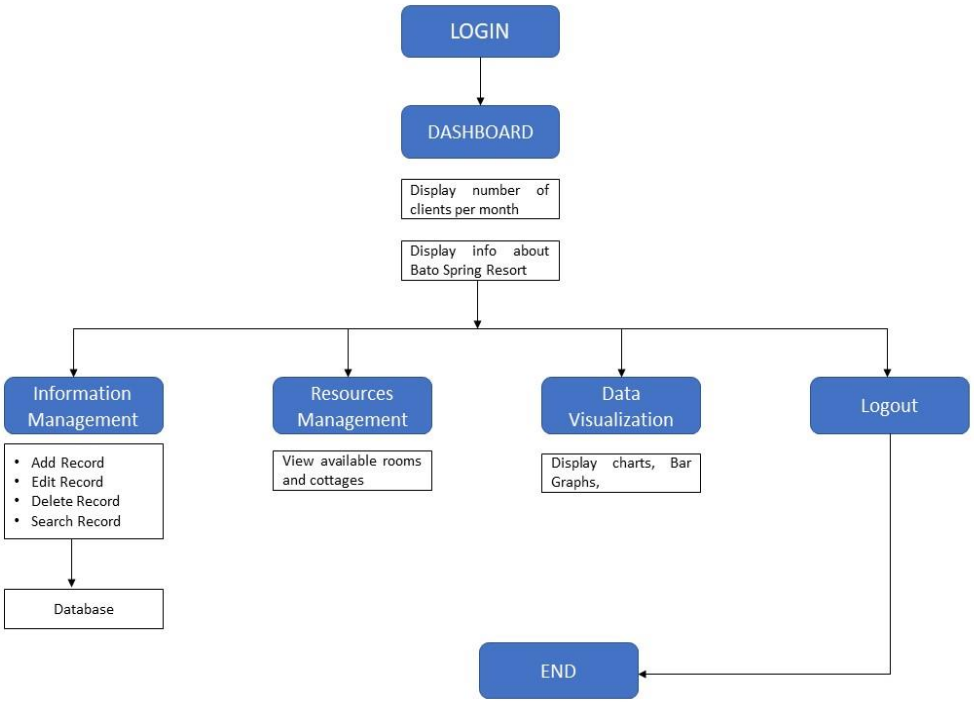


Figure 8. Flowchart

Use Case Diagram

Figure 9 illustrates use-case interaction between two types of users: the User and the Admin within a resort management system. The User has limited access, mainly focused on basic functions such as Registering and viewing available rooms or cottages, allowing guests or clients to check accommodations and create an account. On the other hand, the admin has broader control and system privileges. The admin can add, edit, and delete records, view data visualizations, monitor room and cottage availability, and view and analyze the predicted number of clients to support planning and decision-making. Overall, the diagram clearly shows the difference in access levels, where users interact with basic services while administrators manage and analyze system data.

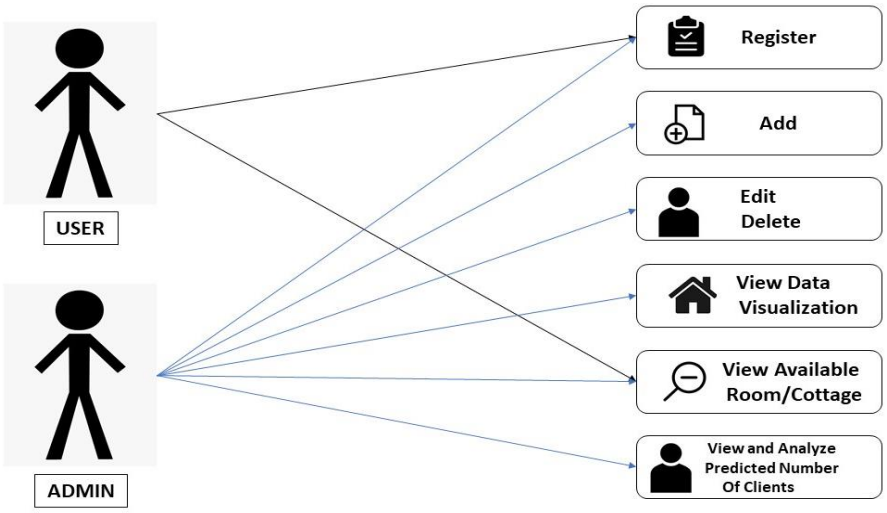


Figure 9. Use Case Diagram

Materials

The following materials have been identified as essential components for the study and was utilized in various stages of the research process.

Programming Language

Python. In this study, the Python has been utilized as the primary programming language. Python is a programming language that is object-oriented, functional, high-level language., and procedural (imperative). It is reflective and structured, and comes with automatic garbage collection. Additionally, it is a language with dynamic typing that is recognized for its syntax that resembles English. Python has many uses, including web, desktop, mobile, and data science.

Libraries. Listed below is a collection of related Python Libraries that helped in the development of the model.

TensorFlow. An open-source software library is called TensorFlow created for extensive artificial intelligence applications and numerical computations. It was first released to the public in 2015 by the Google Brain team. Through shared programmatic analogies, TensorFlow integrates a range of models for machine learning and deep learning and algorithms and makes their use easier.

Pandas. In this investigation, pandas are utilized. Data science, machine learning and data analysis are the most common applications for the open-source Python library Pandas. Data sets are worked with using a Python package called Pandas. Its features include data analysis, cleaning, exploration, and manipulation. The model developed for this study made use of this feature. The Pandas library was used in this study to manipulate data. Utilized for data manipulation and analysis. This library was essential for cleaning, organizing, and preprocessing historical resort data such as guest bookings, occupancy rates, and resource usage.

NumPy. An array object with multiple dimensions, a range of derived items (like matrices and masked arrays), and several procedures for fast array operations—such as sorting, selecting, I/O, discrete Fourier transforms, mathematical, logical, shape manipulation, random simulation, fundamental statistical operations, basic linear algebra, and much more are offered by the NumPy library. The NumPy library is used in this study to carry out mathematical operations and functions associated with model development.

Matplotlib. In visualizing data, validation, and performance graphs of the model, Matplotlib is used since it is a Python library used for generating graphs and charts. It encompasses a submodule called pyplot that eases the process of plotting by offering options to regulate line designs, font properties, adjust axes format, and more. It supports an extensive range of diagrams and charts such as histograms, bar graphs, power spectra, error charts, and more.

Keras. This study made use of Kera; To implement neural networks, an advanced deep learning API was created by Google. It is utilized to build neural networks and is written in Python and data augmentation easier to implement. Additionally, multiple backend neural network computation is supported. The Kera's are employed to enhance the dataset to keep the model from overfitting.

Software. The study's software was listed in this section. A brief description of each program's particular use in the study is provided. Usually, software is kept on an external long-term memory device, like a magnetic diskette or hard drive. Software is the collective term for programs, scripts, and applications that run on a device.

HTML/CSS. The common markup language used to create the structure and content of webpages is HTML (Hypertext Markup Language). It describes the headings, paragraphs, images, links, and other elements and tags that comprise the structure of a web page. The presentation and formatting of an HTML document are described using a style sheet language known as CSS (Cascading Style Sheets). It describes the design, color scheme, typefaces, and other visual components of a webpage. HTML and CSS are used to create the structure and style of the web portal. The various components of the user interface, including buttons, forms, and containers, defined using HTML. On the other hand, these elements' colors, typography, spacing, and general layout would all be controlled by CSS.

JavaScript. A popular programming language that makes dynamic and interactive web page behavior possible is JavaScript. It was utilized by the developers to add functionality, manipulate HTML elements, handle user interactions, and communicate with servers asynchronously. It enhanced the user experience by adding interactivity and responsiveness to the website. It can be utilized for features like image sliders, form validations, dynamic content loading, and client-side data processing. JavaScript enabled real-time updates, interactive customization options, and smoother user interactions.

Bootstrap. It is a popular front-end framework with pre-designed and responsive CSS and JavaScript components. It allowed the developers to create consistent and visually appealing web pages quickly. Bootstrap includes a grid system, responsive navigation bars, buttons, forms, and other UI components. Bootstrap was used to simplify the process of development and guarantee a consistent and mobile-friendly user interface. It provides ready-made components and styles that can be customized to match the branding and design requirements of the web portal

Jupyter Notebook. Jupyter Notebook is an interactive document creation and sharing platform that includes narrative text, equations, live code, and visualizations. Data scientists, machine learning engineers, and other researchers who need to interact with data and code frequently use Jupyter Notebooks. Cells in Jupyter Notebooks can hold either text or code. The code contained in code cells can be run, and the cell displays the results of the code execution. Text cells can be used to write narrative text, equations, and other content. These features could make use as Jupyter Notebook are one of the development environments that was utilized in this study.

Google Colaboratory. A machine learning and data analysis tool called Colaboratory ("Colab" for short) enables you can create a single Google Drive document that contains executable Python code, rich text, charts, images, HTML, and more. Works can be together on projects concurrently by sharing their work with others through a sharable link. The platform is free and web-based, and it offers a pre-set virtual machine environment that includes CPU, GPU, and TPU acceleration. This enables to operate intricate machine learning algorithms without procuring costly hardware or managing complex software installations as Colaboratory was used as their main development environment in this study.

Visual Studio Code. "VS Code," or Visual Studio Code, is an open-source, free, and lightweight coding editor. Many languages are supported by Visual Studio Code, including Python, Java, C++, and JavaScript. Visual Studio Code integrates the user-friendliness of a source code editor featuring robust debugging, code completion, and developer tools. They edited their Python code and ran Jupyter notebooks using Visual Studio Code.

Project Testing and Evaluation

The developers applied multiple levels of software testing to ensure the quality, accuracy, and reliability of the developed web-based resort management system. These tests are conducted to verify that each module and feature functioned correctly and aligned with the objectives of providing predictive guest volume insights, managing resort resources, and visualizing relevant data.

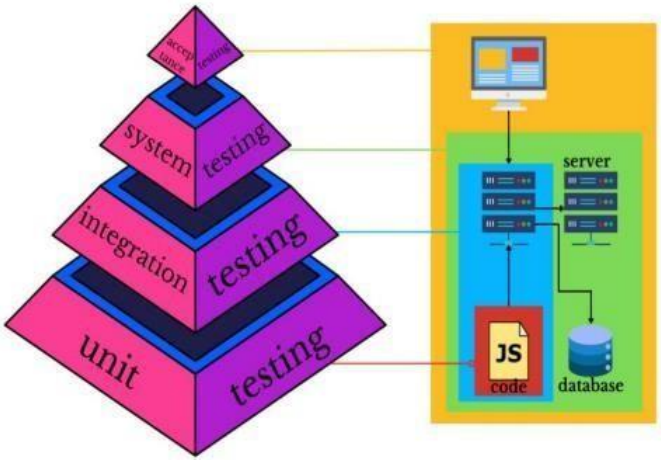


Figure 10. Levels of Software Testing

The figure illustrates the structured approach used to test individual units, module integrations, the entire system, and user acceptability.

Unit Testing. This is the fundamental stage of software testing, concentrating on specific system elements or features. In the resort management system, unit testing was conducted on features such as guest data input, prediction algorithms,
real-time cottage and amenity monitoring, and data graph rendering. These tests ensured each function operated according to specification and produced the expected outputs independently.

Integration Testing. After unit testing, integration testing was performed to verify the interaction between combined modules. For the system, this involved checking the data flow between the guest input module, prediction engine, dashboard, and visualizations. For instance, once guest data is submitted, it must seamlessly feed into the prediction module and display real-time outputs on the dashboard.

System Testing. System testing verified that the entire, integrated system complied with the specified requirements. This involved testing the end-to-end user experience of resort staff, from logging in, inputting guest data, viewing predictions, to monitoring available cottages and amenities. The goal was to confirm that the system delivered accurate information, timely updates, and intuitive navigation across all features.

Acceptance Testing. This final phase evaluated whether the system fulfilled the expectations of the end-users—resort administrators and staff. Test scenarios included simulating real-time guest entries, assessing the accuracy of guest volume predictions, and verifying the clarity of trend graphs and resource availability. The successful completion of acceptance testing indicated that the system was ready for deployment and capable of supporting the resort's operational and strategic needs.

Evaluation Procedure

To ensure that the developed web-based resort management system meets the expected standards of quality, functionality, and user satisfaction, the evaluation was conducted based on the ISO 25010 software product quality model. This international standard defines eight key characteristics that provide thorough framework for evaluating the quality of software. These characteristics include: Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility, and Safety, as illustrated in Figure 10.

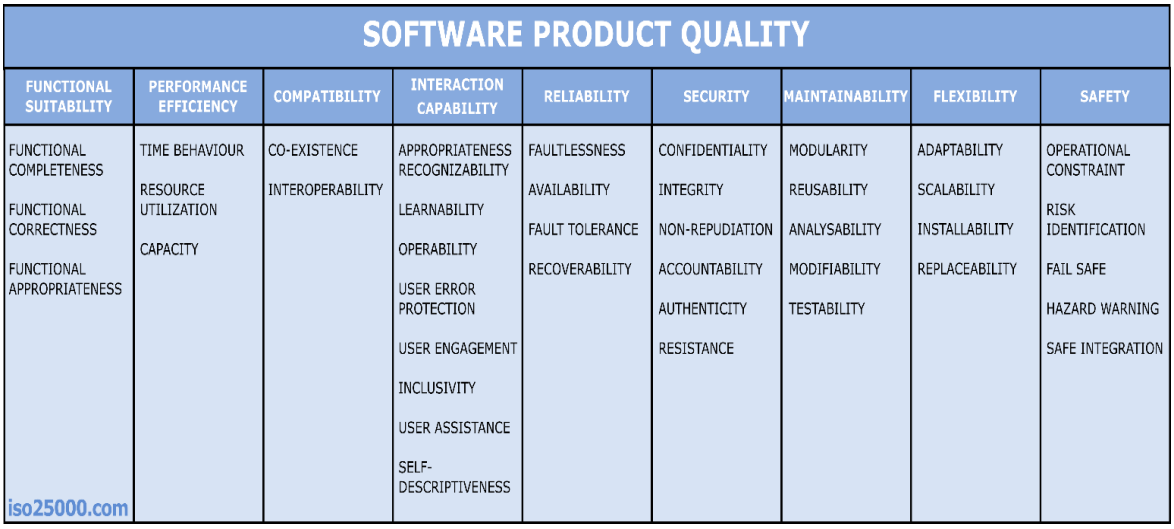


Figure 11. ISO 25010

The system was evaluated through beta testing involving both target end-users (resort staff and administrators) and IT professionals. The evaluators are provided with a structured questionnaire based on the ISO 25010 model, SA stands for Strongly Agree, A for Agree, D for Disagree, and SD for Strongly Disagree on a 4-point Likert scale.

Functional Suitability. Evaluated whether the system meets the functional requirements of Bato Spring Resort, particularly its capability to accurately predict guest volume, track cottage and amenity usage, and generate meaningful visual reports. The feedback confirmed that the system effectively addressed the resort's core operational needs.

Performance Efficiency. Measured the responsiveness and stability of the system under different usage scenarios. The evaluation focused on data processing speed, predictive model responsiveness, and dashboard load time.

Compatibility. Evaluated the system's compatibility with a range of gadgets, such as tablets, laptops, and desktop computers, and web browsers, such as Chrome, Firefox, and Edge.

Interaction Capability. Focused on how user-friendly and intuitive the interface was for resort staff, who may have varying levels of technical knowledge. The system's clear navigation, accessible layout, and visual clarity of reports received high marks from both users and IT experts.

Reliability. Tested the system's ability to perform its intended functions over time without failures. The reliability evaluation included scenarios such as repeated data entry, guest record management, and real-time prediction, all of which are completed successfully without system crashes or data loss.

Security. Reviewed how well the system protects data and prevents unauthorized access. The resort management system includes login authentication and secure access controls, which are evaluated positively for maintaining user privacy and data integrity.

Maintainability. Examined how easily it is possible to update or change the system. Developers and IT specialists assessed the clarity of the code structure, documentation, and database design, concluding that the system could be maintained and improved efficiently over time.

Flexibility. Evaluated the system's the capacity to adjust to changes in operational requirements or technological environments. The web-based design allows for scalability supporting increased data volume, additional resort amenities, or future expansion to other resort branches.

Safety. Assessed how well the system ensures safe operation and prevents potential risks. Safety evaluation focused on risk identification, operational constraints, and fail-safe mechanisms. The system includes automatic data backup, alert notifications for abnormal operations, and safe data integration processes, minimizing hazards such as data loss, system malfunction, or operational delays.

Model Evaluation

Several statistical metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2), used to evaluate the performance of the developed predictive model in this study. These evaluation metrics are commonly used in regression tasks to evaluate how well a model predicts numerical results relative to actual values.

Mean Squared Error

Mean squared error is the average squared difference between the actual and predicted values. Larger errors are penalized more severely due to the squaring process, making it a sensitive indicator of model performance when large deviations are undesirable. Better model accuracy is indicated by a lower MSE. The MSE is mathematically expressed as:

$$\mathbf{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where y_i is the number of observations, n is the predicted value, and y^i i is the actual value.

Root Mean Squared Error (RMSE)

The square root of the MSE is called the Root Mean Squared Error. It makes the error value more comprehensible in real-world situations by returning it to the same scale as the original data. Similar to MSE, improved predictive performance is implied by a lower RMSE. The formula is:

$$\mathbf{RMSE} = \sqrt{\mathbf{MSE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Mean Absolute Error (MAE)

Mean Absolute Error determines the average size of the errors in a set of predictions without considering the direction of the errors. Since it is a linear score, every individual difference has an equal weight. When outliers are not a significant concern, MAE is especially helpful. The formula is:

$$\mathbf{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

A lower MAE indicates that, on average, the predictions are near the actual values.

R-squared (R^2)

The percentage of the dependent variable's variance that can be predicted from the independent variables is shown by the R-squared metric, also known as the coefficient of determination. On a scale of 0 to 1, values nearer 1 indicate that the model explains a significant amount of the variance. It is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where the actual values' mean is represented by $\bar{y}$. A perfect prediction is indicated by an R2 value of 1, whereas a value of 0 indicates that the model does not explain any of the variability.

Dataset

The dataset utilized in this study is a synthetic dataset specifically generated to simulate the operations and guest activity at Bato Spring Resort. It includes a wide range

of relevant information necessary for training the prediction model and creating visualizations. The dataset captures various aspects of guest behavior and operational factors, featuring fields such as the date of visit, number of guests, age range of guests, season or occasion (including holidays), check-in and check-out times, number of rooms or cottages rented, and the type of room or cottage selected.

Additional important attributes are also incorporated to better reflect real-world scenarios, ensuring the dataset closely mimics actual resort operations. Although synthetic, the data is designed to replicate realistic patterns and variations based on typical guest activities observed at the resort.

Since this dataset is artificially generated rather than collected from actual guests, there are no ethical concerns regarding personal data privacy. However, care was taken to guarantee that the dataset remains comprehensive and diverse, enabling the model to learn meaningful patterns for forecasting guest volume and optimizing resource management.

Data Preprocessing

Before the synthetic dataset can be used for training prediction models and generating visualizations, several preprocessing techniques should be applied to enhance data quality and ensure model readiness. First, Simple Imputer can be used to handle missing values by swapping them out for either the mean or the most frequent value, depending on the nature of each column, ensuring that no critical information is lost during model training. To address any potential class imbalance in the dataset.

Oversampling to be applied by duplicating minority instances to create a more balanced distribution, improving model performance and stability. Lastly, Train Test Split is used to separate the dataset into subsets for testing and training, typically with an 80/20 ratio. This prevents overfitting and allows for a more accurate assessment of the model's predictive capabilities by ensuring that it is evaluated using data that has not yet been seen. The stratify parameter are also be considered during the split to maintain consistent class proportions across both sets. Together, these preprocessing steps prepare the dataset for effective machine learning application, ensuring data completeness, balance, and robust evaluation.

Model Development

To predict guest volume based on the operational data from the synthetic dataset, a regression model is employed for training. First, the dataset is preprocessed using the discussed techniques, ensuring that missing values are handled**,** class imbalances are addressed**,** and the data is properly split into training and testing sets. The number of guests is the target variable, and the chosen features—such as the date of visit, season, number of rooms rented, and guest demographics—serve as input variables. To capture intricate connections between the target variable and the features, regression algorithms like Decision Tree, Random Forest, and XGBoost are used. A loss function like Mean Squared Error (MSE), which gauges the discrepancy between the anticipated and actual guest counts, is minimized by the model during training.

Optimization Algorithm: Linear Programming for Resource Utilization

After predicting the number of guests using the trained regression model, Linear Programming (LP) is applied to optimize resource allocation, such as the number of staff members, rooms, and cottages required to efficiently meet guest demand.

LP is a mathematical technique that finds the optimal result (e.g., maximum efficiency or minimum cost) within a given set of constraints.

$$\text{Minimize:} \quad Z = c_1x_1 + c_2x_2 + c_3x_3 + \ldots + c_nx_n$$

The objective function in this study aims to minimize the total operational cost while ensuring that the predicted number of guests are fully accommodated. The basic LP model can be expressed as:

Where Z is the total cost, c_1 represents the cost per unit of resource (e.g., cost per staff member, per room, per cottage), and x_1 is the number of each resource to allocate. Constraints are formulated based on operational requirements, such as:

$$a_1x_1 + a_2x_2 + a_3x_3 + \ldots + a_nx_n \geq \text{Predicted Number of Guests}$$

Where a_1 denotes the capacity of each resource (e.g., one room can accommodate four guests). Additional constraints may include minimum staffing levels, room availability, and seasonal limitations. Solving the LP problem is yielding the optimal number of staff, rooms, and cottages needed for the given guest forecast, ensuring resources are neither over-allocated (which increases costs) nor under-allocated (which reduces service quality). In this study, Python's scipy. optimize. linprog function can be utilized to efficiently solve the Linear Programming model, allowing the optimization process to be seamlessly integrated into the system's workflow for real-time resource allocation and decision-making

Hardware

Table 4. Hardware Specification

	Minimum	Recommended
Processor	AMD A6 7480 @ 3.8ghz	Ryzen 5 5600g @ 3.9ghz
Memory	8 Gigabytes	64 Gigabytes
Hard Disk Drive	128 Gigabytes	1 Terabyte
System Type	3 Operating System X 32 based processor	64-bit Operating System X 64 based processor

Desktop Computer, the stated hardware specification was utilized. In order to ensure replicability and comparability of results, future researchers are advised to utilize the minimum and recommended specifications or improved hardware specifications when conducting similar studies.

Population of the Study

IT specialists and users made up the study's population who actively participated in evaluating and testing the developed system. These individuals are selected because they represent the intended users and stakeholders of the system, and their feedback is essential in assessing the system’s functionality, usability, and overall effectiveness. Table 5 presents the categories and number of respondents involved in the evaluation process

Table 5. 53 Respondents are directly involved in using or analyzing the system's key features. Their insights provided critical input toward identifying strengths and areas where the functionality and design of the system could be improved. The assessment also benefited from the perspectives of IT professionals who are knowledgeable in software quality standards and web systems.

Table 5. Respondents of the Research Project

CATEGORY OF RESPONDENTS	SAMPLE
User	50
IT Expert	3
Total	53

Sampling Design

In this study, a non-probability sampling method called purposeful sampling was employed. The researchers purposefully chose participants based on their availability and system relevance. The respondents are chosen for their direct involvement in resort operations and their familiarity with the functions of the developed system. Using this method, the researchers are able to get insightful input from people who are most likely to use or evaluate the system successfully. A total of 53 participants are selected, comprising users (50) and (3) IT professionals. Their combined experience and knowledge of the resort's day-to-day activities and technical systems made them suitable evaluators of the platform's functionality and usability.

Data Collection Instrument

To gather data, the researchers used both survey questionnaires and interviews. The survey was divided into two sections. The first section focused on the general user experience and measured how well the system addressed the needs of its users in terms of design, ease of use, responsiveness, and functionality. First Section which includes 30 items, focused on evaluating if the program fulfilled the requirements of the user and used the Technology Acceptance Model (TAM) to examine the answer of the respondents.

The second section was based on the ISO 25010 software quality model, covering eight key software attributes: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. These are evaluated using the following choices on a four-point Likert scale: Strongly Agree (SA), Agree (A), Disagree (D), and Strongly Disagree (SD). Both printed and digital (Google Forms) versions of the survey are distributed to accommodate respondent preferences.

Semi-structured interviews are conducted with select participants to gain deeper insights and specific feedback about the system's strengths, weaknesses, and areas for future improvement. This multi-faceted data collection strategy ensured a comprehensive evaluation of the web-based system and helped validate its effectiveness in enhancing resort operations and user satisfaction.

Statistical Treatment

Purposive sampling was used in this study to collect pertinent data from a chosen group of people who are directly involved in the management and assessment of the web.

system for managing resorts. The population was divided into important respondent groups, such as resort employees, management staff, and IT specialists, to guarantee a thorough representation of the resort's user base. This approach allowed for the effective assessment of the system from both administrative and technical perspectives.

To evaluate the usability, functionality, and performance of the developed system, the researchers employed a Likert Scale as the primary statistical tool. The Likert Scale consisted of a range of answers, ranging from Strongly Agree (5) to Strongly Disagree (1), providing a structured and measurable method for capturing user feedback.

Table 6. Likert Scale Technique for ISO

Rating	Mean Range	Interpretation
4	3.25 — 4.00	Strongly Agree
3	2.50 — 3.24	Agree
2	1.75 — 2.49	Disagree
1	1.0 — 1.74	Strongly Disagree

The researchers strategically utilized the Likert Scale as the primary method for evaluating user satisfaction with the developed system. This survey approach provided a structured mechanism for collecting user feedback across various dimensions of the system, such as design, usability, performance, and functionality. By offering respondents the Likert Scale has a range of Strongly Disagree to Strongly Agree enabled the capture of nuanced opinions and varying levels of agreement. This method allowed the research team to gain a deeper understanding of user sentiments and experiences, especially in relation to how the system met their operational needs and enhanced resort management processes. Through this technique, the developers are able to assess not only the general acceptance of the system but also specific areas of improvement, making the Likert Scale an essential tool in analyzing user perception and guiding further system enhancements.

Technology Acceptance Model (TAM), This framework supports the end-users to understand and accept the developed system. Davis introduced the Technology Acceptance Model (TAM) in 1989, cited by Thompson in 2019 provide a foundation for comprehending the elements effecting prospective user’s acceptance of computer systems consumers. In term of perceived usefulness, consumers perceptions of how benefits of some technologies improve system output or satisfy their requirements,

However, consumers perception of how simple and easy to use the technology is in terms of interaction and navigation is what is meant by perceived of use.

The researchers used this method to determine the extent to which the system is easy to use and how each module is functionable and reliable. And if the system is useful. This study collected responses from Bato Spring Resort's management and IT personnel to evaluate how the system's machine learning features contribute to predicting guest.

Volume and improving operational decisions. The results of this evaluation are essential in determining users' behavioral intention to use the system and in ensuring that the interface design, performance, and system benefits meet user expectations

7 categories of TAM; Perceived of Usefulness, perceived of ease of Use, Personal Innovation, Social Influence, Information Awareness, Perceived Cost, and Openness to Adopt. To determine the outcomes of the respondent's evaluation, each item is scored on a 4- point Likert scale. The findings aid in identifying important variables affecting user satisfaction and adoption. This analysis offers valuable insights for improving system design and implementation strategies.

Table 7. 4- Point Verbal Interpretation Scale for TAM

Rating	Interpretation
4	Strongly Agree
3	Agree
2	Disagree
1	Strongly Disagree

The 4-point Verbal Interpretation Scale for TAM. The respondents evaluate the system using this rating to interpret the user's acceptance level and assessing user's attitudes towards adopting new technologies.

The scale shows the interpretation for each rating: (4) Strongly Agree, (3) Agree, (2) Disagree, (1) Strongly Disagree.

Table 8 depicts the 4-Point Likert Scale used for the Technology Acceptance Model (TAM). The scale is designed to capture respondents' perceptions and level of agreement with statements regarding the system, guiding them in expressing the degree of acceptance or willingness to use the technology. The composite mean was calculated by averaging responses across key TAM constructs, including perceived usefulness, perceived ease of use, innovation, social influence, information awareness, perceived cost, and intention to adopt the technology.

This approach allows for a quantitative assessment of overall user acceptance, providing insight into how respondents view the system in terms of functionality, accessibility, and practicality. By analyzing the composite mean, researchers can identify areas of strength—such as features that are highly valued by users—and potential areas for improvement, such as components perceived as costly or difficult to adopt. For example, a composite mean between 3.25 and 4.0 indicates "Strongly Agree," 2.50 to 3.24 indicates "Agree," 1.75 to 2.49 indicates "Disagree," and 1.0 to 1.74 indicates "Strongly Disagree", providing a clear framework for interpreting respondents' perceptions.

Table 8. 4 – Point Likert Scale for TAM

Rating	Composite Mean	Interpretation
4	3.25 – 4.0	Strongly Agree
3	2.50 – 3.24	Agree
2	1.75 – 2.49	Disagree
1	1.0 – 1.74	Strongly Disagree

CHAPTER 4

RESULT AND DISCUSSION

This chapter presents the outcomes of the study and discusses how the applied methods contributed to the accomplishment of its objectives.

Results by Objectives of the Study

Figure 12 presents the user's first view of the Bato Spring system, showcasing a clean, modern, and welcoming homepage designed to immediately engage users while providing intuitive access to the system's core features and navigation tools. As the initial point of interaction, this landing page is essential in forming the user's first impression of the system and the resort it represents. In addition to navigation, the homepage features clear and well-positioned call-to-action buttons, such as options to view resort resources and plan a visit. These buttons guide users toward key system functions and encourage interaction without overwhelming them with excessive options.

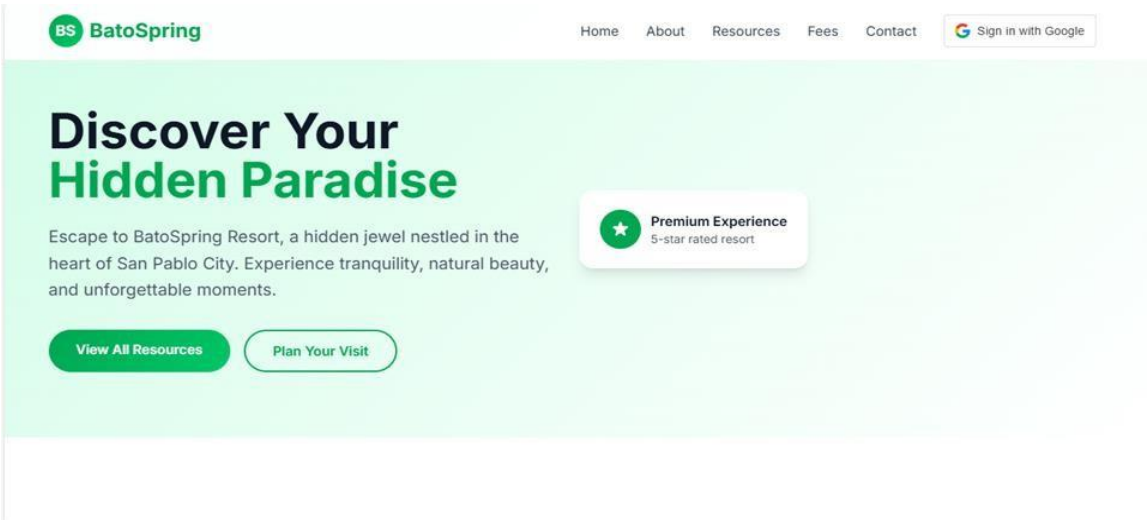


Figure 12. Landing Page

Figure 13 allows users to easily access the system by signing in with their Google account. When the user clicks the “Sign in with Google” button, a window appears where they can choose or add their account. This feature provides a fast, convenient, and secure way for users to log in to the to Spring Resort system.

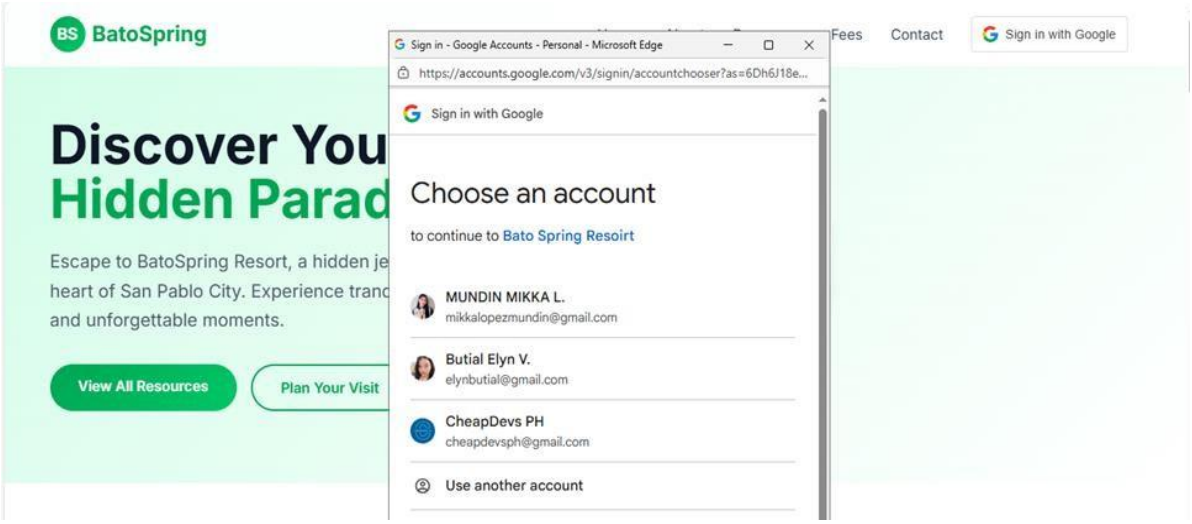


Figure 13. Login Module

Figure 14 serves as the main control panel of the Bato Spring Resort system, where users can view their active bookings, total visits, notifications, and loyalty points.

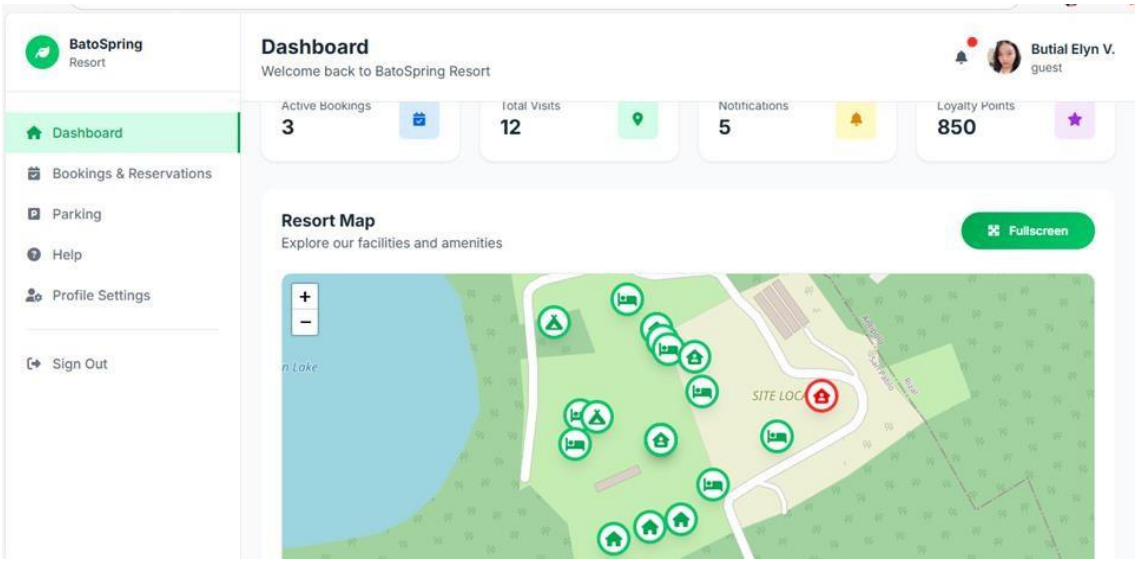


Figure 14. Dashboard

Figure 15 presents the Profile Settings page enables users to securely manage and update their account information within the BatoSpring Resort system. It shows essential details such as the user's profile photo, full name, email address, phone number, address, and assigned role (Guest). The sidebar provides easy navigation to other system features, while the save and cancel buttons allow users to confirm or discard changes. This page helps maintain accurate user data, supports smooth communication, and enhances overall system usability and user experience.

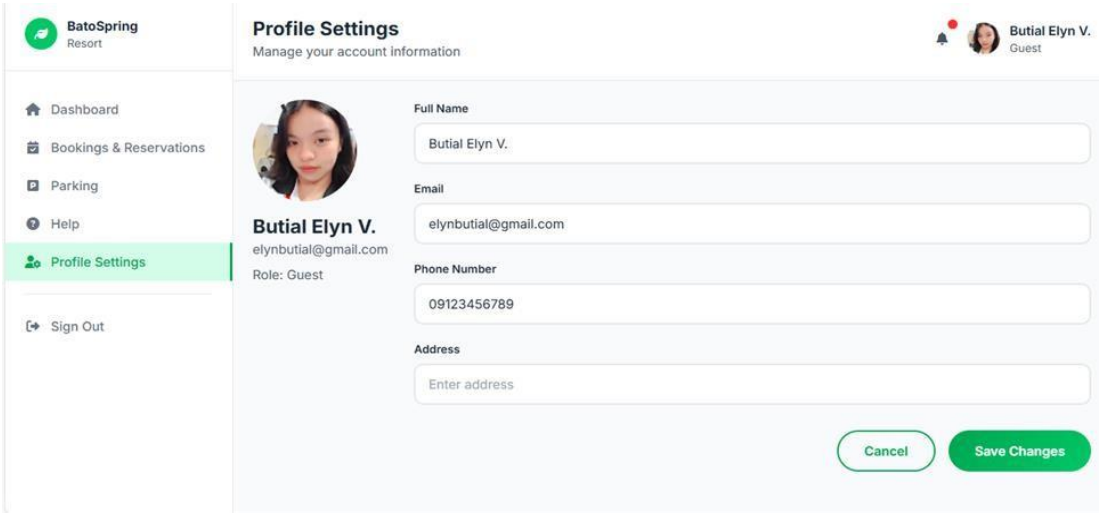


Figure 15. Profile Settings (User)

Figure 16 The Predictions and Analytics page of the Bato Spring Resort Admin Panel showcases a sophisticated, AI-driven dashboard designed to support data-informed decision-making by providing administrators with a clear, comprehensive, and visually intuitive overview of the resort's overall performance. This page integrates advanced analytics and predictive insights to highlight key operational trends, guest behavior patterns, and resource utilization metrics, enabling management to better understand both historical performance and future projections. Through well-organized data visualizations, real-time reports, and actionable forecasts, administrators can identify peak seasons, anticipate guest demand, optimize staffing and inventory, and improve service delivery.

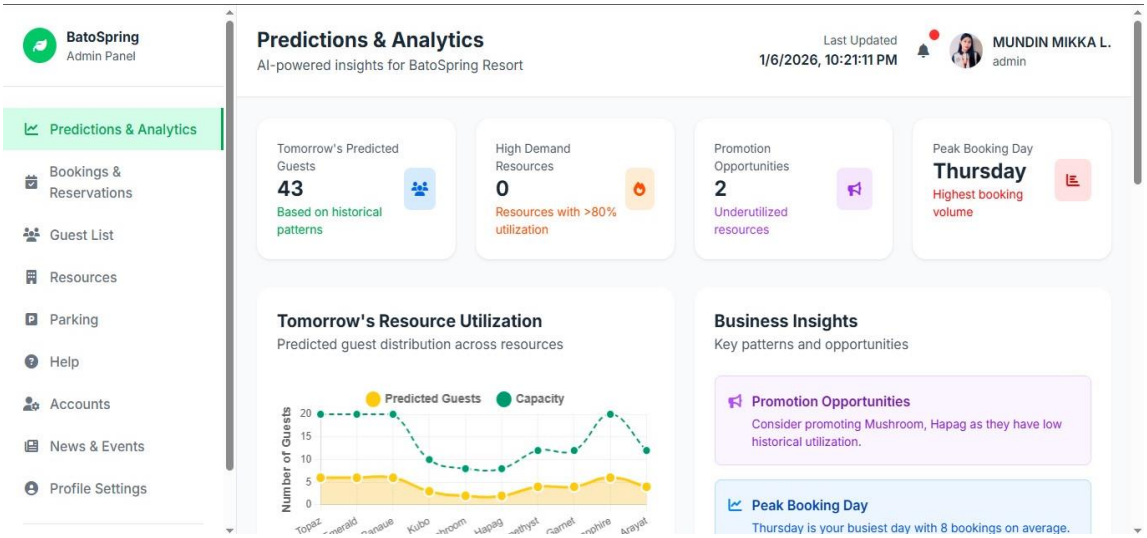


Figure 16. Prediction and Analytics

This figure 17 enables administrators to efficiently manage and track all resort bookings and reservations. It displays key information such as the total number of bookings, pending approvals, confirmed reservations, and cancellations. The table below provides detailed records for each guest, including their name, reserved resource, check-in and check-out dates, payment amount, and current status. Administrators can filter bookings by status, resource, or date, and export the data for reporting or recordkeeping purposes

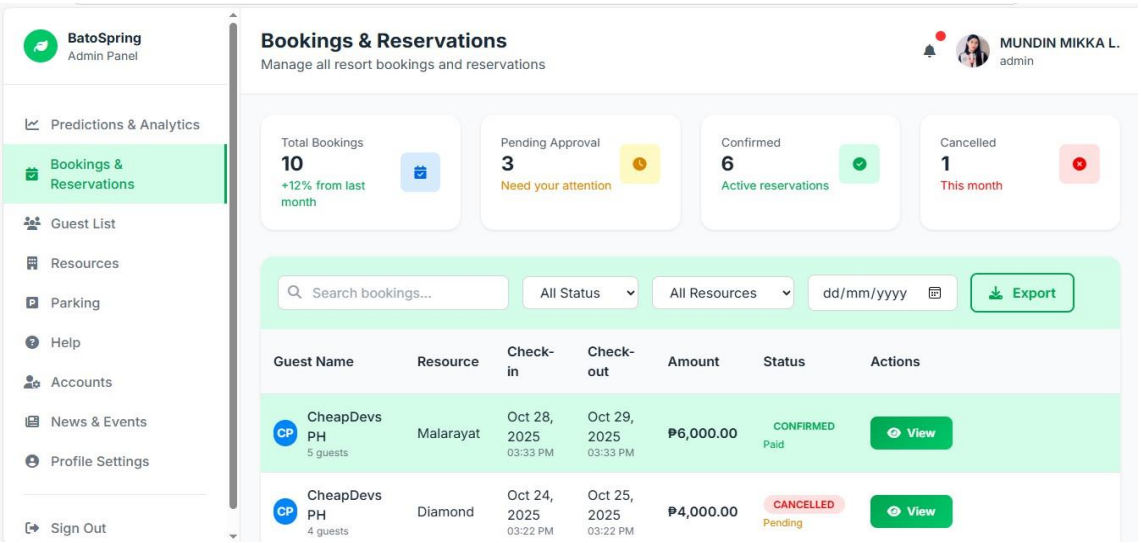


Figure 17. Bookings and Reservation(admin)

This figure 18 allows administrators to manage and view all resort guests, displaying details such as guest name, email, phone number, number of visits, last visit date, and total amount spent. It also provides options to filter guest data, track visit frequency, and export guest information for record management.

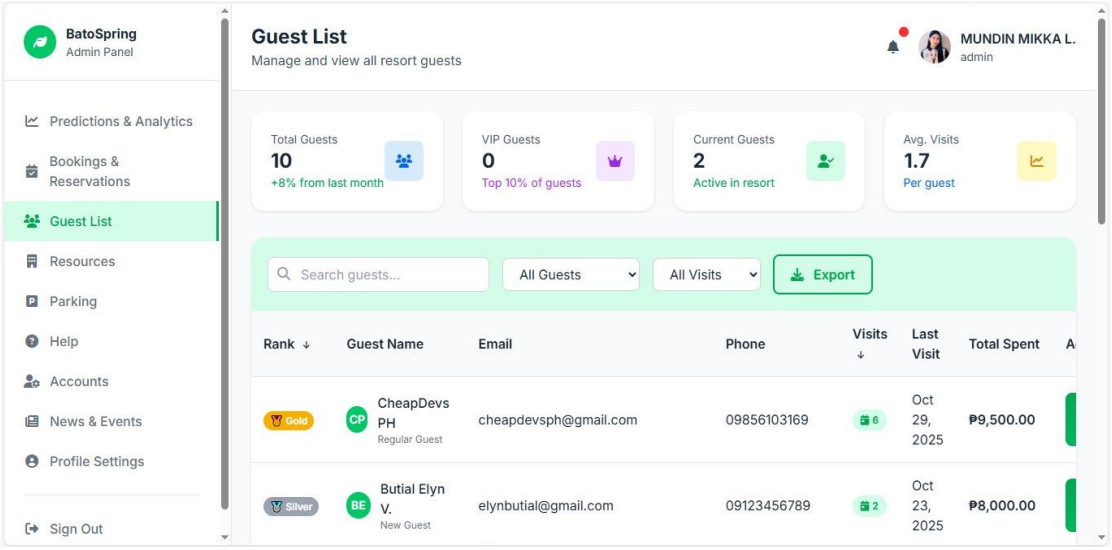


Figure 18. Guest list (Admin)

Figure 19 illustrates the Resources Management page of the Bato Spring Resort Admin Panel, which is intended to give administrators a centralized, effective interface for managing all resort resources and facilities in a systematic and organized manner, ensuring smooth daily operations and accurate monitoring of availability. The page features a clean and user-friendly layout, beginning with a summary section at the top that displays key resource statistics, including the total number of resources, available resources ready for booking, currently occupied resources, and those under maintenance, allowing administrators to quickly assess the overall resource status at a glance. On the left side, a navigation menu offers easy access to other administrative modules, with the Resources section clearly highlighted to indicate the active page.

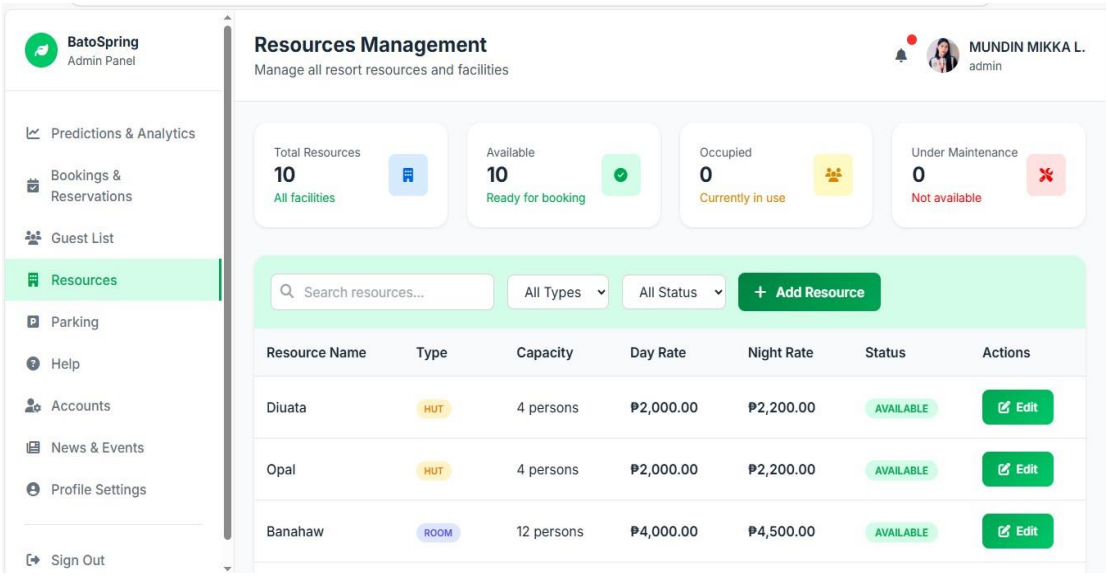


Figure 19. Resources management

The figure 20 Bato Spring Resort System's Accounts Management page gives administrators a centralized interface for monitoring and controlling all user and admin accounts. Through this page, administrators can add new accounts, update existing account information, reset passwords, and deactivate or remove accounts as necessary.

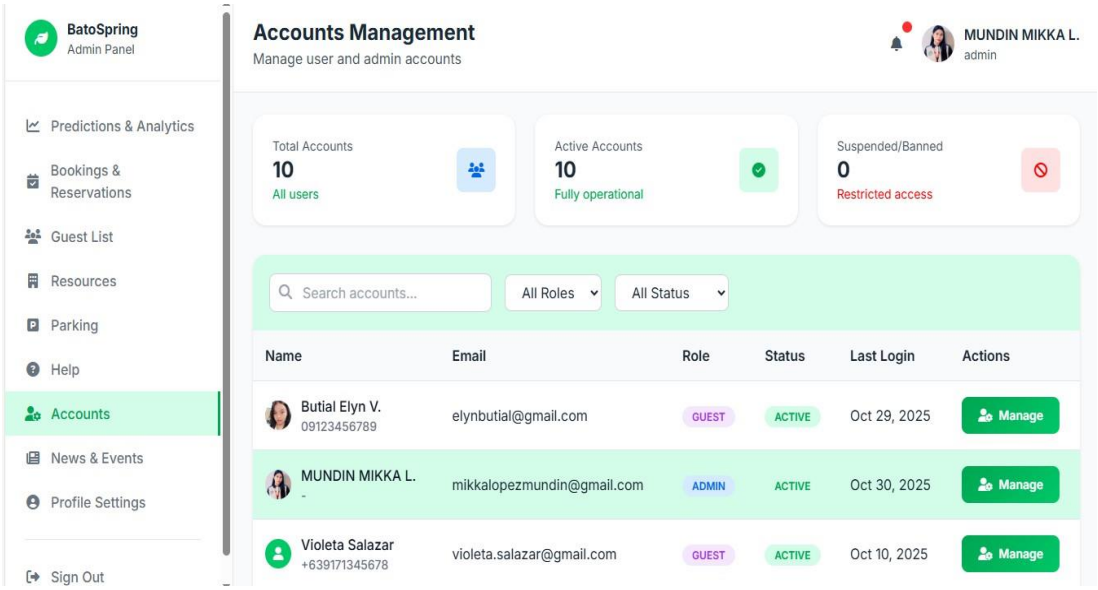


Figure 20. Accounts

The system transforms raw booking data from Bato Spring Resort into meaningful insights through data analysis, machine learning prediction, and business intelligence reporting to enhance resort management and decision-making.

System Functionalities

Data Analysis. Enables the understanding of booking patterns, revenue trends, and customer behavior.

Machine Learning Integration. Uses predictive algorithms to estimate future guest numbers and optimize resource planning.

Forecasting. Predicts guest volumes for upcoming days to support effective workforce scheduling and resource allocation.

Figure 22 shows the sequential process of the Bato Spring Resort Data Analytics and Prediction System. The process begins with API Data, where information on bookings, resources, and users is collected directly from the resort's management system. Next, during Data Processing, the collected data is cleaned and enhanced through feature engineering to ensure quality and consistency. The Analysis phase follows, where data is visualized and analyzed statistically to identify trends and patterns in guest behavior and resource usage.

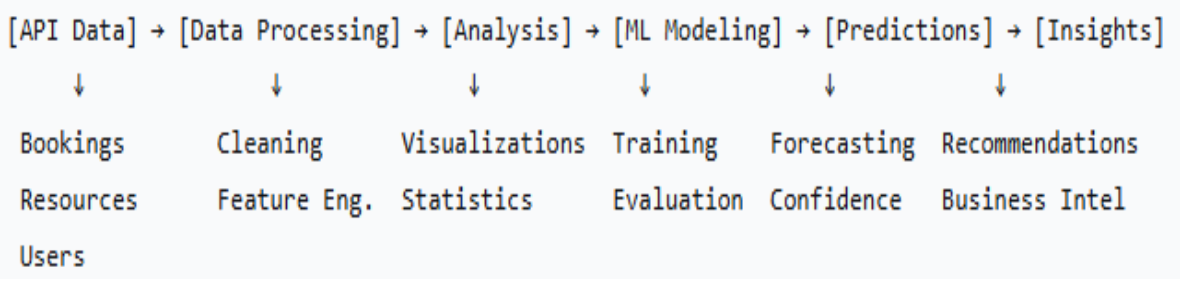


Figure 22. System Architecture and Workflow

Machine Learning Model Implementation

Because of its high accuracy, capacity to handle missing data, and ease of interpretation, the Random Forest Algorithm is used by the system to forecast the number of guests at Bato Spring Resort. It is useful for predicting guest numbers and optimizing resource management because it combines several decision trees to generate accurate and consistent predictions. To guarantee prediction accuracy and dependability, the Random Forest model's performance was evaluated using a number of evaluation metrics. Random Forest is an ensemble learning technique that builds numerous decision trees during the training phase and produces predictions by combining the outcomes of these trees. This process aids in minimizing overfitting and enhancing the overall stability of the model.

Table 9. Model Performance and Evaluation

Metric	Value	Description
MAE	.8500	Mean Absolute Error (guests)
MSE	1.2300	Mean Squared Error
RMSE	1.1100	Root Mean Squared Error
R^2	0.8200	R-squared
MAPE	12.5000	Mean Absolute Percentage Error
RMSE % of Mean	18.3000	RMSE as Percentage of Mean

With a Mean Absolute Error (MAE) of 0.85 and a Root Mean Squared Error (RMSE) of 1.11, which show that the predictions differ very little from the actual guest numbers, the model demonstrated strong predictive performance. The model has a high degree of explanatory power, as evidenced by the R2 value of 0.82, which shows that it accurately captures 82% of the variance in guest attendance. Furthermore, a Mean Absolute Percentage Error (MAPE) of 12.5% and RMSE as a percentage of the mean (18.3%) show that prediction errors stay within reasonable bounds for realistic forecasting, indicating that the model is accurate and reliable in a variety of data scenarios.

These results confirm that the model provides reliable and accurate forecasts for Bato Spring Resort, supporting better planning and operational decision-making. By accurately predicting guest volume, the system can optimize resource allocation, staff scheduling, and inventory management, thereby reducing operational costs, minimizing wastage, and improving service efficiency. Furthermore, the model allows management to anticipate peak periods and low-demand intervals, enabling proactive planning for promotions, maintenance, or staffing adjustments.

The strong performance also highlights the effectiveness of machine learning integration in enhancing resort management, offering data-driven insights that can improve strategic decision-making, operational efficiency, and overall guest satisfaction. In the long term, this predictive capability can serve as a foundation for continuous improvement, supporting adaptive strategies that respond to seasonal trends, special events, or unexpected fluctuations in guest arrivals. Overall, the model demonstrates that leveraging advanced analytics and predictive modeling can transform resort operations into a more efficient, responsive, and customer-focused system.

Test and evaluate the web-based system using the ISO 25010

The respondents evaluated the system positively, with an overall mean score of 3.63, reflecting a strong perception of the system's quality. Among the assessed attributes, Performance Efficiency (M = 3.78) and Portability (M = 3.75) received the highest ratings, indicating that users found the system to be highly efficient, responsive, and adaptable across different devices or environments. Reliability and Security, while slightly lower at M = 3.50, are still classified as "Strongly Agree," suggesting that these areas perform well but may benefit from further enhancements to meet the highest standards of system dependability and protection.

The low standard deviation (SD = 0.42) highlights the consistency of participants' perceptions, showing that the majority of users share similar positive views regarding the system's quality attributes. These results indicate that the system is well-designed, user-friendly, and capable of supporting operational needs, while also providing insight into areas where further optimization—such as enhancing security protocols and reliability under heavy use—could strengthen overall system performance.

Table 10. ISO 25010 Result

Characteristics	Mean	SD	Interpretation
Functional Suitability	3.56	0.58	Strongly Agree
Performance Efficiency	3.78	0.39	Strongly Agree
Compatibility	3.67	0.39	Strongly Agree
Usability	3.59	0.3	Strongly Agree
Reliability	3.50	0.39	Strongly Agree
Security	3.50	0.58	Strongly Agree
Maintainability	3.67	0.39	Strongly Agree
Portability	3.75	0.39	Strongly Agree
Safety	3.67	0.32	Strongly Agree
Overall Mean	**3.63**	**0.42**	**Strongly Agree**

Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), Two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

Test and evaluate the web-based system using the Technology Acceptance Model

(TAM)

Table 11 shows that users perceive the system positively**,** with mean scores ranging from 2.78 to 2.96**,** indicating a general agreement on its usefulness and effectiveness**.** The system is particularly valued for enhancing resource management efficiency and improving the accuracy of guest volume forecasting, which are critical aspects of resort operations. Additional benefits highlighted by users include enhanced staff productivity (mean = 2.78), access to valuable information for resource allocation decisions (mean = 2.80), and improved overall resort operations through the integration of machine learning (mean = 2.84)**.**

The low standard deviation and standard error further suggest that user feedback is consistent and reliable, reflecting a shared positive perception across respondents. These results indicate that the system is regarded as a significant asset for operational efficiency, accurate forecasting, and informed decision-making within Bato Spring Resort. Moreover, the integration of data-driven insights through machine learning enables administrators to proactively plan resources and optimize scheduling.

Table 11. Results of Perceived Usefulness

Item	Mean	Standard Deviation	Interpretation
The system helps me perform resource management tasks more efficiently.	2.96	1.01	Agree
The system improves the accuracy of guest volume forecasting.	2.96	0.90	Agree
Using the system enhances my productivity at work.	2.78	0.76	Agree
The system provides useful information for decision-making regarding resource allocation.	2.80	0.83	Agree
The integration of machine learning in the system improves resort operations.	2.84	0.87	Agree
Composite Mean	**2.87**	**0.87**	**Agree**

Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 2) – Disagree (1.76 – 1.50), and one (1) – Strongly Disagree (1.0 – 1.75)

Table 12 indicates that respondents generally perceive the system as easy to use, with mean scores ranging from 2.66 to 2.80, corresponding to the “Agree” category. Users report that the system is highly navigable, requiring minimal effort to interact with its features, and they appreciate its intuitive and user-friendly interface. Many respondents noted that the system’s functions are clearly labeled and logically organized, which facilitates task completion and reduces the learning curve. Additionally, users indicated that they could easily remember how to use the system over time, reflecting its design for long-term usability.

The low standard deviations further demonstrate consistent positive evaluations, showing that respondents share similar experiences regarding its accessibility and ease of use. These results suggest that the system successfully achieves high usability standards, promoting efficient interaction, reducing potential user frustration, and supporting widespread adoption among staff and guests.

Table 12. Result Perceived Ease of Use

Item	Mean	Standard Deviation	Interpretation
Interacting with the system does not require a lot of effort.	2.78	0.84	Agree
The system interface is user-friendly and well-organized.	2.78	0.82	Agree
I can complete my tasks quickly when using the system.	2.66	0.85	Agree
The system’s functions are clear and understandable.	2.80	0.81	Agree
I can easily remember how to use the system even after some time.	2.74	0.80	Agree
Overall Mean	**2.75**	**0.82**	**Agree**

Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

Table 13 presents respondents' assessment of their behavioral intention to use the system and their openness toward new technologies. The results show mean scores ranging from 2.64 to 2.74, all interpreted as "Agree," with an overall mean of 2.68, indicating a generally positive attitude toward adopting and experimenting with innovative systems. Respondents demonstrate a willingness to learn, adapt, and engage with new technology, reflecting their readiness to incorporate modern tools into their workflow.

The consistent mean scores and relatively low standard deviations suggest that these positive perceptions are shared widely among users, highlighting a collective openness to technological advancement. This readiness is significant for the successful implementation and adoption of the Bato Spring Resort system, as it implies that users are likely to actively engage with the system, explore its features, and integrate it into their daily operations. Overall, the findings underscore that the system not only meets usability and functionality expectations but also aligns well with users' attitudes toward innovation and technology adoption, which is essential for long-term system sustainability and effectiveness.

Table 13. Result Personal Innovation

Item	Mean	Standard Deviation	Interpretation
I like experimenting with new technologies and systems.	2.66	0.83	Agree
I am prepared to learn and adapt to new technologies.	2.66	0.87	Agree
I am open to adopting innovative technologies.	2.64	0.83	Agree
I enjoy trying out new technologies like this system.	2.74	0.92	Agree
Overall Mean	**2.68**	0.86	**Agree**

Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), wo (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

Table 14 presents respondents' evaluation of social influences on their system usage. The mean scores range from 2.28 to 2.80, with an overall average of 2.60, indicating general agreement that social factors positively affect engagement with the system. Respondents report feeling supported by their social environment, emphasizing that encouragement from colleagues and peers influences their readiness to adopt and actively use the system.

Although the perception of social pressure to use the system is slightly lower, with a mean of 2.28, it still reflects that peer influence contributes to shaping user behavior. These findings suggest that social support, peer acceptance, and collaborative endorsement are significant factors in promoting system engagement. In practice, fostering a workplace culture where staff encourage one another, share knowledge, and model effective system usage can enhance adoption rates, compliance, and overall satisfaction with the Bato Spring Resort system. This aligns with the Social Influence construct in the Technology Acceptance Model (TAM),

Table 14. Result Social Influence

Item	Mean	Standard Deviation	Interpretation
People around me support the use of this system.	2.66	0.85	Agree
Seeing that others benefit from the system makes me want to try it too.	2.80	0.86	Agree
I feel social pressure to use the system because others expect me to.	2.28	0.96	Agree
My friends and colleagues think it's a good idea to use the system.	2.64	0.83	Agree
Composite Mean	**2.60**	**0.88**	**Agree**

Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

Table 15 presents respondents' assessment of the perceived costs and efforts associated with using the system. The mean scores range from 2.16 to 2.28, with an overall average of 2.21, indicating a consensus of "Disagree." This suggests that most respondents do not perceive the system as expensive or burdensome. Users generally disagree that system maintenance, upgrades, or operation would require substantial financial resources, significant time, or excessive effort, nor do they feel that more economical alternatives would provide a clear advantage.

Respondents report that there are no significant expenditures or effort-related barriers that might hinder system usage. These findings imply that the Bato Spring Resort system is cost-effective, efficient, and user-friendly, minimizing potential obstacles to adoption. By combining affordability with ease of use, the system is positioned to encourage wider acceptance, reduce resistance, and promote sustainable integration into daily operations. This perception of low cost and effort further supports the system's practicality and accessibility for both staff and administrators.

Table 15. Result Information Awareness

Item	Mean	Standard Deviation	Interpretation
I know what the system is designed to do.	2.64	0.90	Agree
I am aware of the features and functions of the system.	2.74	0.90	Agree
I understand how the system helps in managing Bato Spring's resources.	2.06	0.79	Disagree
I have enough information about how to use the system properly.	2.58	0.88	Agree
Overall Mean	**2.51**	**0.87**	**Agree**

Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1)

Table 16 presents respondents' assessment of their willingness to adopt and use the system. The mean scores range from 2.22 to 2.76, reflecting a mixed perception of sustained system adoption. Most respondents' express disagreement with statements regarding regular use, recommending the system to others, and future usage, indicating some hesitation or reluctance toward fully integrating the system into their routine activities.

The mean score of 2.76 for the statement on regular use suggests that some respondents recognize the potential value of the system and may be open to using it under favorable conditions. Overall, these results indicate that while the system is generally perceived as useful and easy to use, additional measures such as enhancing system functionality, providing user training, or fostering positive user experiences may be necessary to increase acceptance, encourage continuous use, and promote advocacy among users. Addressing these factors can help ensure that the system achieves long-term adoption and maximizes its benefits for Bato Spring Resort operation

Table 16. Result Perceived Cost

Item	Mean	Standard Deviation	Interpretation
Maintaining or upgrading the system might be costly.	2.24	1.00	Disagree
Using the system may take too much of my time and energy.	2.28	1.13	Disagree
I believe there are more affordable alternatives to this system	2.16	0.84	Disagree
Using the system may involve expenses that concern me.	2.16	0.89	Disagree
Overall Mean	**2.21**	**0.97**	**Disagree**

Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1)

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The mean score of 2.76 for the statement on regular use suggests that some respondents recognize the potential value of the system and may be open to using it under favorable conditions. Overall, these results indicate that while the system is generally perceived as useful and easy to use, additional measures such as enhancing system functionality, providing user training, or fostering positive user experiences may be necessary to increase acceptance, encourage continuous use, and promote advocacy among users. Addressing these factors can help ensure that the system achieves long-term adoption and maximizes its benefits for Bato Spring Resort operation

Table 17. Result Willingness to Adopt

Item	Mean	Standard Deviation	Interpretation
I am ready to use this system regularly.	2.22	0.93	Disagree
I recommend this system to others.	2.22	0.93	Disagree
I plan on continuing to use the system in the future.	2.24	1.00	Disagree
I am open to use this system regularly. *(Duplicate item)*	2.76	0.89	Agree
Overall Mean	**2.36**	**0.94**	**Disagree**

Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

The Technology Acceptance Model (TAM) summary shows an overall mean of 2.57 with a standard deviation of 0.89, interpreted as “Agree,” indicating generally positive user perceptions of the system. Respondents largely agree that the system is useful, easy to use, and aligns with their personal openness to adopting new technologies, suggesting favorable attitudes toward initial adoption. Additionally, social influence and information awareness are rated positively, highlighting the supportive role of peers and the accessibility of relevant system information in encouraging usage.

Perceived Cost and Openness to Adoption received comparatively lower ratings, indicating concerns about potential financial or resource-related burdens and some hesitation regarding sustained long-term use. These findings suggest that while users accept and recognize the system’s benefits, additional measures such as cost justification, user training, and strategies to reinforce long-term engagement may be necessary to increase continued adoption and advocacy. Overall, the results demonstrate that the Bato Spring Resort system is well-received, but addressing cost perceptions and fostering sustained usage will be critical for maximizing its impact on operational efficiency and decision-making.

Table 18. Summary of TAM

Item	Mean	Standard Deviation	Interpretation
Perceived Usefulness	2.87	0.87	Agree
Perceived Ease of Use	2.75	0.82	Agree
Personal Innovation	2.68	0.86	Agree
Social Influence	2.60	0.88	Agree
Information Awareness	2.51	0.87	Agree
Perceived Cost	2.21	0.97	Disagree
Willingness to Adopt	2.36	0.94	Disagree
Overall Mean	**2.57**	**0.89**	**Agree**

Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

Project Testing and Debugging

A. Unit Testing

Unit testing was conducted to verify the accuracy and functionality of each individual component of the Bato Spring Resource Utilization Optimization and Management System Integrating Machine Learning Algorithm before integrating them into the overall system. This process ensured that every unit, such as the login module, data input form, resource management function, machine learning prediction model, and data visualization component, performed according to the specified requirements. Each test focused on detecting and correcting errors in coding logic, data validation, and output generation.

Table 19. Conducted test case for Unit Testing

Unit Test ID	Steps	Expected Results	Actual Results	Remarks
Guest Data	password, then click “Login.”	credentials” error message.	expected.	Passed
Entry Test	Input complete guest information (date, number of guests, contact info), then click “Submit.”	Guest data saved and displayed in database table.	Data successfully saved and reflected in records.	Passed
Input Validation Test	Leave one or more input fields empty, then click “Submit.”	System prompts “All fields are required.”	Validation alert appeared correctly.	Passed
Resource Reservation Test	Reserve a cottage or room under Resource Management module.	Resource status changes from “Available” to “Occupied.”	Status updated successfully in real time.	Passed
Machine Learning Prediction Test	Upload test dataset and run ML prediction function.	System generates numerical guest volume prediction.	Prediction displayed accurately on dashboard.	Passed

B. Integration Testing

Integration testing was performed after the successful completion of the unit testing phase to verify that the individual modules of the Bato Spring Resource Utilization Optimization Management System integrating a Machine Learning Algorithm functioned correctly when combined. This stage focused on testing the interaction between interconnected components, including the login authentication, data input and storage, resource management, machine learning prediction, and data visualization modules. The primary objective was to ensure that data flowed accurately across the system—for example, confirming that guest information entered in the data input module was properly stored in the database and subsequently reflected in the prediction and visualization modules.

Integration testing was performed after the successful completion of the unit testing phase to verify that the individual modules of the Bato Spring Resource Utilization Optimization Management System integrating a Machine Learning Algorithm functioned correctly when combined. This stage focused on testing the interaction between interconnected components, including the login authentication, data input and storage, resource management, machine learning prediction, and data visualization modules. The primary objective was to ensure that data flowed accurately across the system—for example, confirming that guest information entered in the data input module was properly stored in the database and subsequently reflected in the prediction and visualization modules.

Integration testing also verified that the web interface communicated seamlessly with the machine learning model and database, ensuring no errors, data loss, or miscommunication occurred between components.

Table 20. Test case conducted for Integration Testing

Test ID	Steps	Expect of Results	Actual Results	Remarks
Login Dashboard Integration Test	Log in using valid username and password, then access the dashboard.	The system successfully verifies credentials and loads the user dashboard.	Dashboard displayed properly with user information and menu options.	Passed
Guest Data Input Test	Input guest data in the data entry form and submit. Then check database and dashboard.	Guest data is stored in the database and reflected in the dashboard summary.	Data saved correctly and appeared in dashboard and reports.	Passed
Resource Reservation Test	Reserve a resource (cottage or room) after guest data is saved.	The reservation status updates in the resource management module.	Resource status changed from “Available” to “Occupied” in real time.	Passed
Resource Release Test	Release a reserved resource and refresh dashboard.	Resource status updates back to “Available” and dashboard reflects change.	Status updated successfully and dashboard synchronized.	Passed
Admin Update Consistency Test	Log in as admin, update a record, and check data consistency across all modules.	Updated information reflects consistently in database, dashboard, and reports.	All changes synchronized across modules without error.	Passed
Unauthorized Access Prevention Test	Log out and attempt to access dashboard via direct URL.	System should deny access and redirect to login page.	Access denied and redirected properly to login page.	Passed

C. System Testing

System testing was conducted after the successful completion of integration testing to ensure that the Bato Spring Resource Utilization Optimization and Management System Integrating Machine Learning Algorithm functioned as a fully operational software product. This stage aimed to validate that the entire system met all specified functional and non- functional requirements outlined in the project objectives.

During this phase, the researchers simulated real-world scenarios by performing end-to-end testing of major features, including user authentication, data entry, resource management, machine learning predictions, and data visualization.

Moreover, system testing assisted in pinpointing mistakes, performance problems, and usability issues. By tackling these problems at an early stage, the system became more reliable, user-friendly, and stable. In summary, system testing confirmed that the developed system is operational, dependable, and prepared for deployment at Bato Spring Resort.

Table 21. Test Case for Load Test

Test ID	Steps	Expected Results	Actual Results	Remarks
System Launch and Login Test	Launch the system and log in using authorized credentials.	System loads successfully, validates login, and displays dashboard.	Dashboard loaded correctly without delays.	Passed
Guest Data Input Test	Input new guest information and submit.	Data is saved, stored in the database, and reflected in analytics.	Guest data saved and displayed in real time.	Passed
Resource Reservation Test	Perform resource reservation and verify update on dashboard.	Resource status updates to “Occupied” and dashboard reflects change.	Resource management module updated successfully.	Passed
Machine Learning Prediction Test	Run machine learning model to generate guest volume prediction.	System processes data and displays an accurate prediction output.	Prediction generated accurately and displayed on the dashboard.	Passed
Data Visualization Update Test	View data visualization section after new records are added.	Graphs and analytics update automatically with new data.	Graphs refreshed and displayed updated values correctly.	Passed
Data Update and	Edit or delete guest record	Updated or deleted data	Data synchronized	Passed

CHAPTER 5

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This chapter summarizes the main conclusions of the study, emphasizing its importance and how the findings solve the issues and goals mentioned in the introduction. Additionally, it offers suggestions for future enhancements and promotes more investigation into related study areas.

Summary

The researcher carried out a study titled "Bato Spring Resource Utilization Optimization and Management System Integrating Machine Learning Algorithm" in order to create a web-based system that would enhance operational effectiveness and optimize resource management at Bato Spring Resort. The researcher created a system that combines real-time data monitoring, predictive analytics, and resource optimization to support proactive and informed management after realizing the growing demand for data-driven decision-making in the hospitality sector.

The researcher first focused on designing and developing a web-based system capable of managing and optimizing the resort’s resources. The system includes modules for data collection, analysis, and visualization, enabling resort administrators to monitor guest bookings, track resource usage, and generate actionable insights. By centralizing these functions, the researcher ensured that operational processes are streamlined and that resources could be allocated efficiently according to demand.

The researcher integrated a machine learning model into the system to predict guest volume using historical and real-time data. The predictive model enable management to anticipate guest arrivals and allocate resources such as rooms, personnel, and utilities more effectively. In addition, the researcher incorporated an

optimization algorithm that distributes resources based on forecasted demand, reducing waste, lowering operational costs, and enhancing overall service quality. This integration of prediction and optimization allows resort administrators to make proactive, informed decisions rather than relying solely on manual estimations or historical trends.

The researcher conducted a system evaluation using the ISO 25010 software quality model. The evaluation results demonstrated that the system met quality standard was considered reliable, efficient, and user-friendly by its intended users. This indicates that the researcher successfully developed a system that not only fulfills its technical requirements but also provides a practical tool for improving resort operations.

Overall, the researcher achieved the objectives of the study by creating a robust, web- based resource management system that leverages machine learning and optimization techniques. The system provides Bato Spring Resort with a technology-driven solution that enhances operational efficiency, promotes effective resource utilization, and supports sustainable management practices. Through the efforts of the researcher, the study offers a practical framework for data-informed decision-making, enabling resort administrators to respond proactively to fluctuations in guest volume and ultimately improve both service quality and operational performance

Conclusion

Created a web-based system for Bato Spring Resort. The researcher created an effective web-based system that effectively oversees and maximizes the use of resources at Bato Spring Resort. The system improves overall operational efficiency by offering a centralized and organized platform for tracking resource usage, keeping an eye on reservations, and producing actionable insights.

The integration of a machine learning model and optimization algorithm enables accurate prediction of guest volume and effective allocation of resources such as accommodations, personnel, and utilities. This functionality allows resort management to make proactive, data-driven decisions that minimize waste and enhance service quality.

Evaluation results using the ISO/IEC 25010 software quality model confirmed that the system met high standards in terms of functionality, usability, reliability, and performance efficiency. Furthermore, the Technology Acceptance Model (TAM) assessment revealed a high level of user acceptance, indicating that resort administrators and staff perceived the system as useful, easy to use, and beneficial to their daily operations.

Overall, the study proves that integrating machine learning and optimization techniques into resort management systems significantly enhances resource utilization, supports data-driven decision-making, and promotes sustainable operations.

Evaluation of system using the ISO 25010 software quality model

Using the ISO 25010 software quality model confirmed that it meets established standards in terms of Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility, and Safety. The results indicate that the system is technically sound, user-friendly, and practical for resort management purposes.

The usability and acceptance of the system are evaluated using the Technology Acceptance Model (TAM)

Strong user and staff acceptance and satisfaction are found in an evaluation using the Technology Acceptance Model (TAM), indicating the system's usefulness, efficiency,

and practicality. In general, the study emphasizes the potential of Technology-driven approaches to boost resort administration, raise service standards, and maximize resource use.

Recommendation

1. Implementation

As technology evolves, it is imperative for Bato Spring Resort to implement the developed resource utilization optimization and management system fully and ensure its continuous use, as it contributes to enhanced operational efficiency, optimize resource allocation, improve service quality, and support informed, data-driven decision-making by the management. The implementation of the Bato Spring Resource Utilization Optimization and Management System Integrating Machine Learning Algorithm involved the integration of a web-based platform, a machine learning prediction model, and an optimization algorithm to support intelligent resource management. The system was developed using an Agile Scrum methodology, allowing iterative development, testing, and refinement based on user requirements and feedback.

2. Regular System

Regular system updates and maintenance should be conducted to ensure that the machine learning model remains accurate and reliable, allowing the system to continuously adapt to the resort's evolving operational needs and technological advancements, thereby maintaining optimal performance and supporting effective resource management. This system does not provide real-time tracking of guest volume or resource utilization. As a result, management depends largely on experience and

estimation when planning staffing schedules, facility preparation, and daily operations. During peak seasons and holidays, this often leads to overcrowding, delayed services,

and overworked staff. Conversely, during off-peak periods, resources may remain underutilized, causing inefficiencies and unnecessary operational costs.

3.Staff Training

Staff should receive comprehensive training on using the developed system to ensure its smooth integration into daily operations, maximize its effectiveness in optimizing resource utilization, and enable employees to confidently leverage its features for improved decision-making and overall operational efficiency.

Future improvements may involve the systematic expansion of the system to integrate a comprehensive range of additional data sources, including seasonal trends, customer feedback, historical booking patterns, real-time environmental data, and other relevant operational metrics, with the aim of further refining predictive accuracy, enhancing resource planning, supporting informed and proactive decision-making, and ultimately improving the overall efficiency, effectiveness, and strategic management of Bato Spring Resort's operations in a dynamic and evolving business environment

4.Further Research

Future researchers may further enhance this study by exploring the use of advanced machine learning and deep learning models such as Long Short-Term Memory (LSTM), XGBoost, or hybrid forecasting models to improve guest volume prediction accuracy, especially for handling complex seasonal patterns and long-term trends. Comparing these models with the Random Forest algorithm using larger and more diverse datasets may provide deeper insights into model performance and scalability.

Additionally, future studies may integrate real-time data sources such as weather conditions, public holidays, local events, and online booking behavior to enrich the

prediction model and improve demand forecasting. Incorporating Internet of Things (IoT) sensors for monitoring water usage, electricity consumption, and facility occupancy may also strengthen resource optimization and support sustainable resort management.

Furthermore, research may also focus on expanding the system to include financial and billing modules**,** such as dynamic pricing strategies based on predicted demand, cost optimization analysis, and revenue forecasting. Evaluating the system across multiple resorts or tourism establishments could improve its generalizability and adaptability.

Moreover, complementary evaluation frameworks other than ISO/IEC 25010 and the Technology Acceptance Model may be used in future research. These models may not capture organizational and strategic outcomes, despite being useful for evaluating software quality and user acceptance. A more comprehensive understanding of value creation can be obtained by combining the DeLone and McLean Information Systems Success Model, the Balanced Scorecard, and sustainability assessment frameworks.

Finally, in order to thoroughly investigate the long-term impacts of machine learning-based resource management systems on resort operations, future researchers are urged to carry out longitudinal studies. Deeper evaluation of how intelligent systems affect operational effectiveness, cost reduction, guest satisfaction, and environmental sustainability over long implementation periods would be made possible by such studies.

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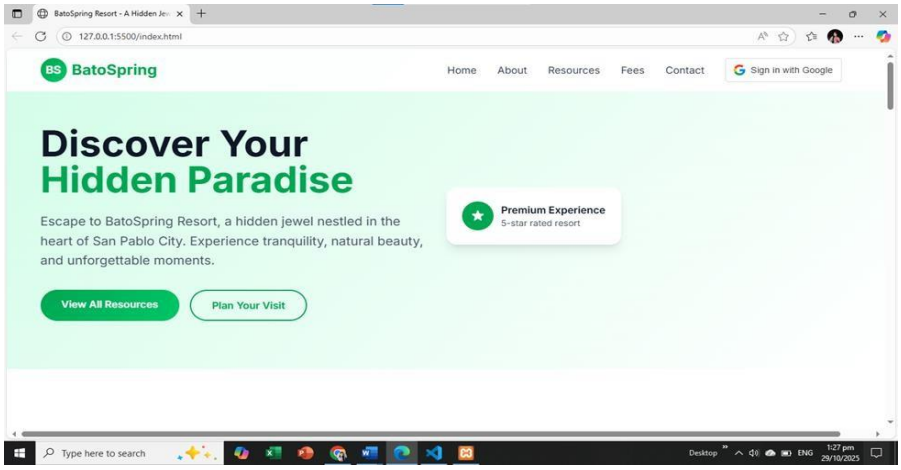
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APPENDIX A.

Output Screenshot

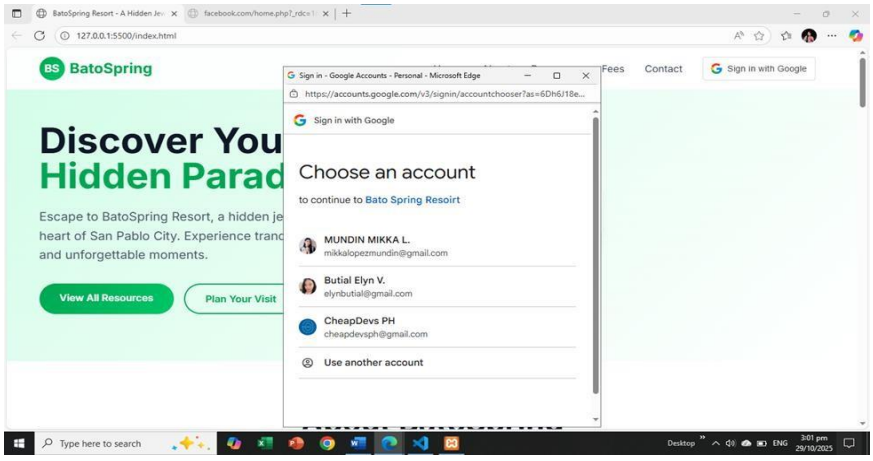
Landing page

It is a webpage made to get visitors to do one specific action, like sign up, book, or learn more.



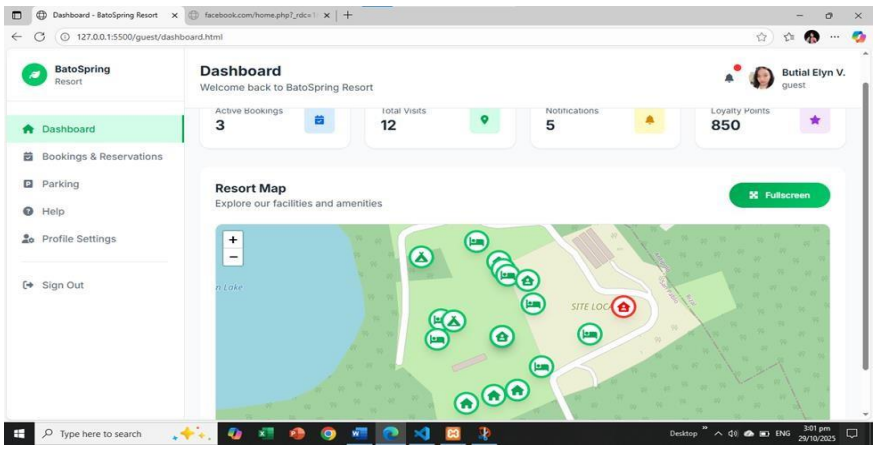
User log in

Log in using Google means signing into a website or app directly with your Google account without creating a separate password.



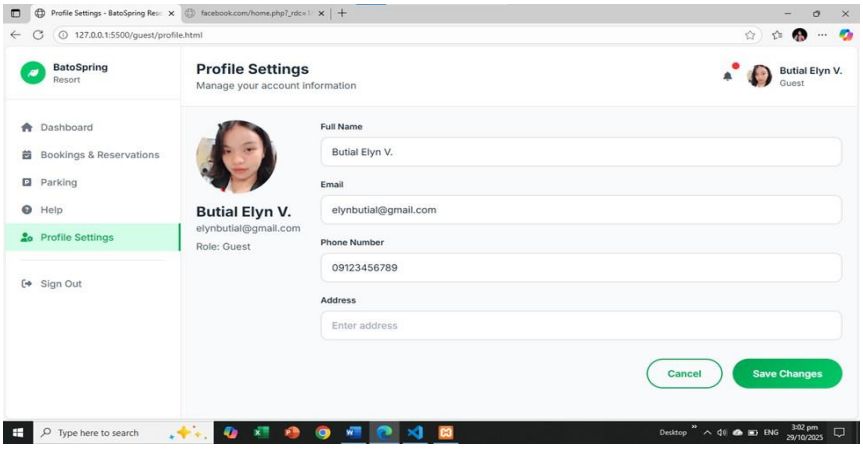
Dashboard User

It is a personalized interface where a user can view and manage their account, activities, and settings on a website or app.



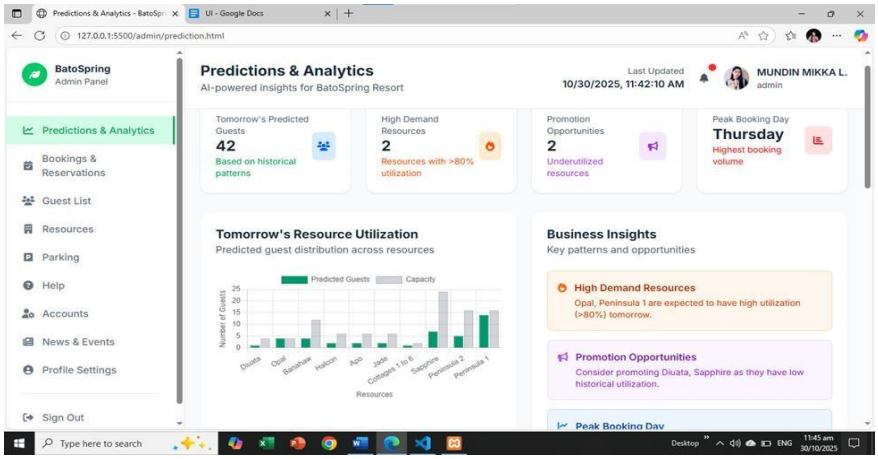
Profile setting User

It is the section where a user can edit personal info, change passwords, and manage account preferences.



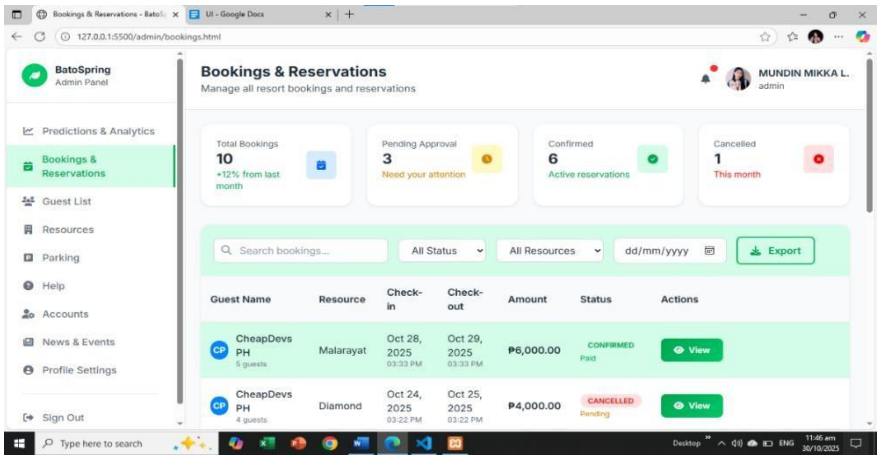
Prediction and analytics

It refers to using data to forecast future trends or outcomes (prediction) and analyze past and present data to gain insights for better decision-making (analytics).



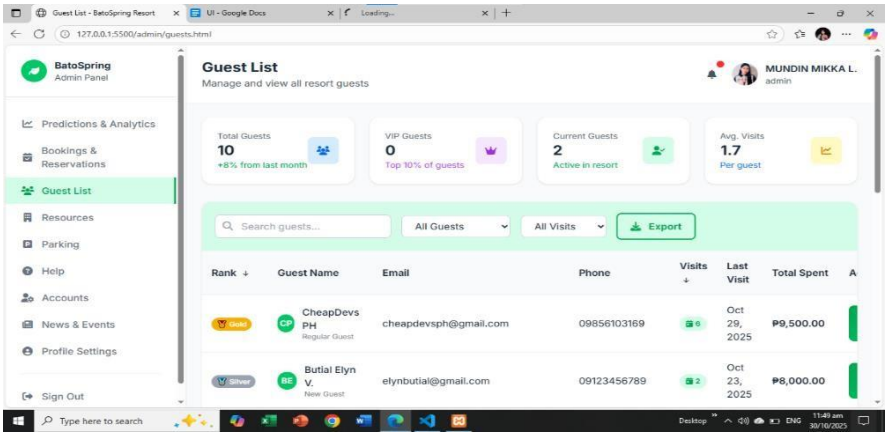
Bookings and reservation

It refers to the system or section where users can reserve rooms, cottages, or other facilities, view their upcoming and past bookings, and manage check-in/check-out dates.

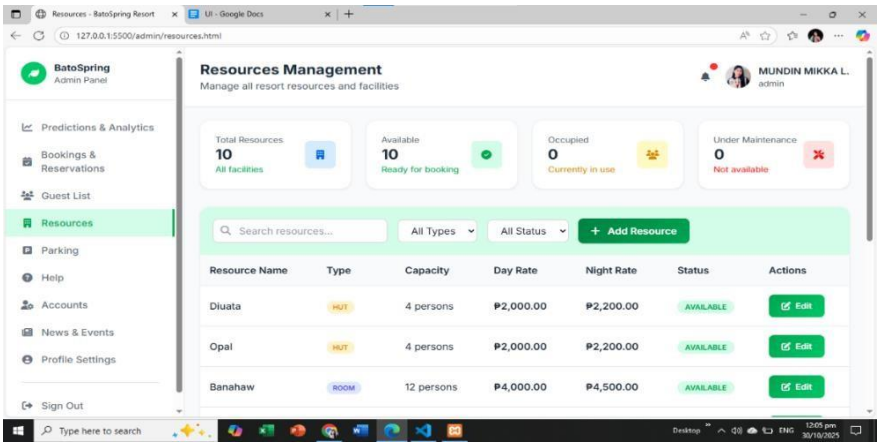


Guest List

It refers to a record or section that displays all visitors or guests who have booked or checked in, including their names, contact details, booking dates, and status. It helps the resort track attendance, manage reservations, and organize guest information efficiently.



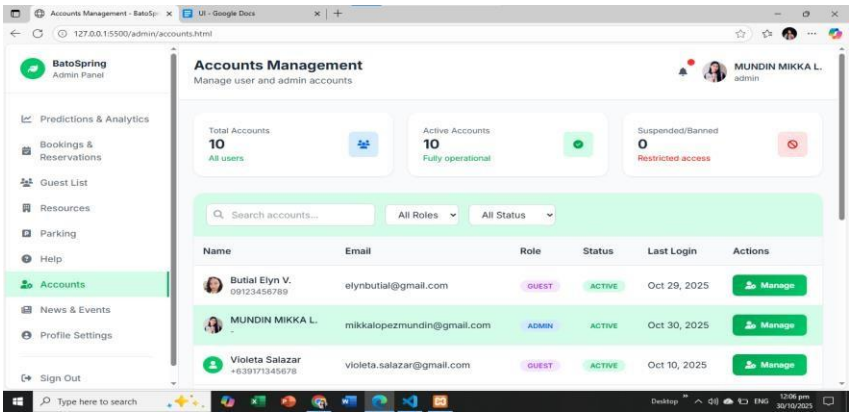
Resources management



This is the process of organizing, tracking, and optimizing the use of a resort resources, such as rooms, cottages, tables, rate, to ensure efficient operations and availability**.**

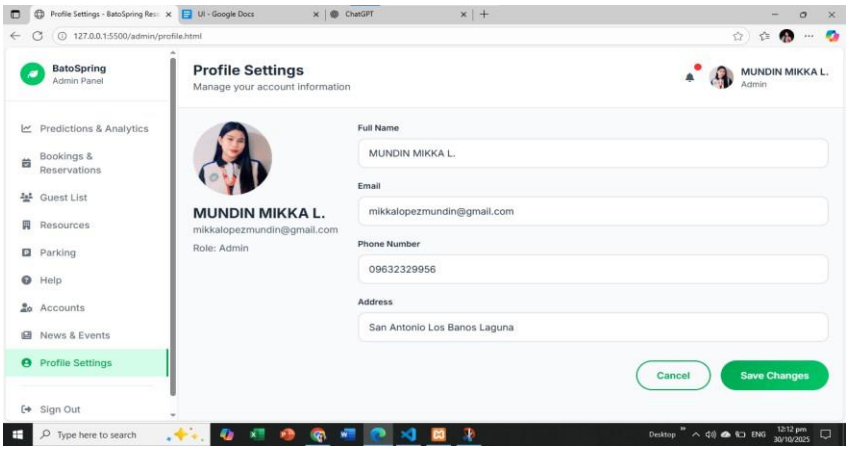
Accounts management

The image shows a modern Accounts Management dashboard with a teal and blue design focused on clarity and usability. This is the process of handling user accounts, including creating, updating, and monitoring user information, access levels, and permissions, to ensure secure and organized account operations within a system.



Profile Settings

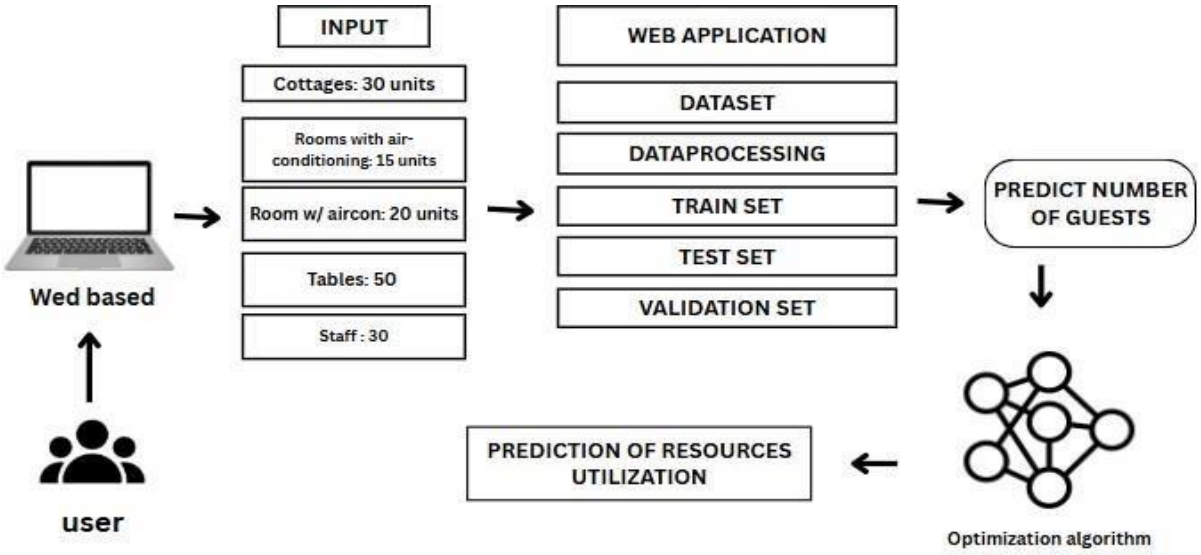
This is the section where the administrator can view and update their personal information, change passwords, and manage system preferences. It may also include options to configure user roles, access permissions, and account security settings, ensuring proper control and customization of the system.



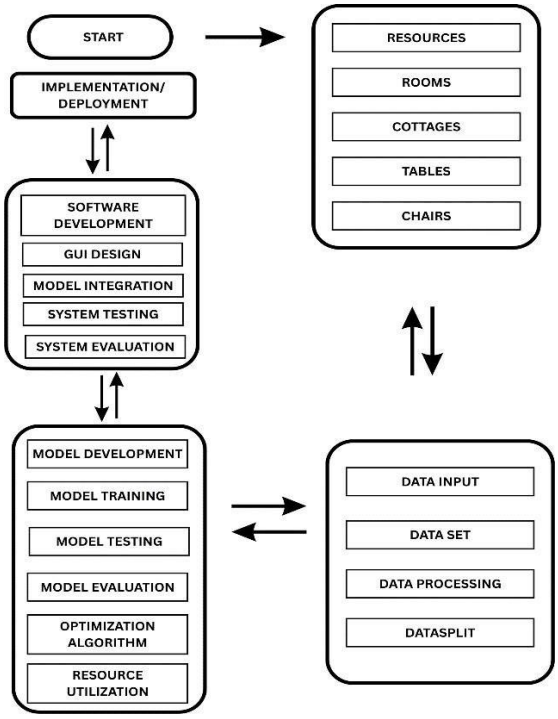
APPENDIX B.

Figures

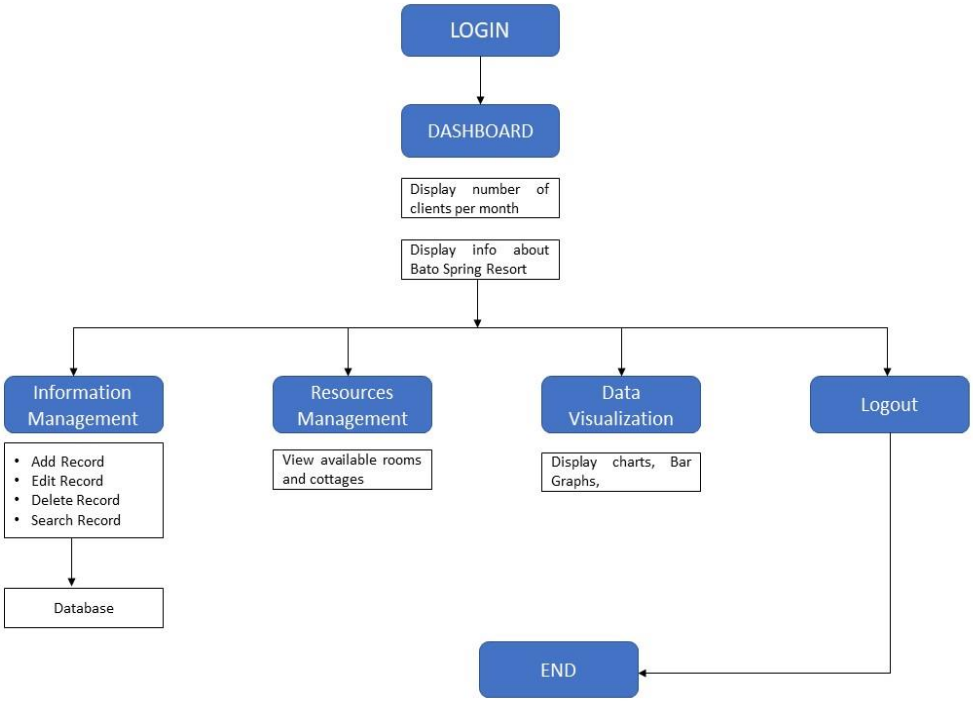
Conceptual Framework



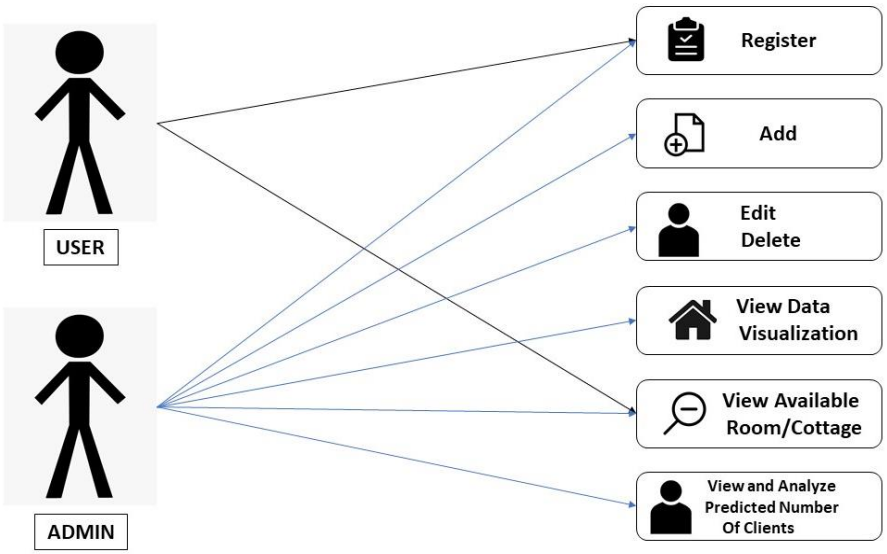
System Architecture



Flowchart



Use Case Diagram



APPENDIX C.

Source Code

```
<?php
namespace App\Models;

use
App\Core\Database;
use
App\Models\Auth;
use PDO;
use PDOException;

class Admin {
  // Profile
  public static function findByApiKey($apiKey, $csrfToken) {
    $db = Database:connect();
    $stmt = $db->prepare ("
      SELECT * FROM users
      WHERE api_key =
      :api_key
        AND csrf_token = :csrf
      AND role = 'admin'
      LIMI
    T 1
    ");
    $stmt->execute ([
      'api_key' =>
      $apiKey,  'csrf' =>
      $csrfToken
    ]);
    return $stmt->fetch (PDO:FETCH_ASSOC);
  }

  public static function updateProfile($userId, $updates) {
    $db = Database:connect();

    // Build dynamic query
    $set = [];
    foreach ($updates as $field => $value) {
      $set [] = "$field = :$field";
    }

    $sql = "UPDATE users SET " . implode (', ', $set) . ", updated_at = NOW () WHERE user_id =
:user_id";
    $stmt = $db->prepare($sql);

    $updates['user_id'] = $userId;
    return $stmt-
    >execute($updates);
  }

  // News
  public static function isDuplicateNews($title, $description) {
    $db = Database:connect();
    $stmt = $db->prepare ("SELECT COUNT (*) FROM news WHERE title = title AND description =
: description");
    $stmt->execute ([
      'title' =>
      trim($title),
      'description' => trim($description)
    ]);
```

```
    return $stmt->fetchColumn() > 0;
  }

  public static function createNews($newsData) {
    $db = Database:connect();

    $sql = "INSERT INTO news
          (user_id, news_id, title, description, image_url, event_date, status, created_at, updated_at)
          VALUES (:user_id, :news_id, :title, :description, :image_url, :event_date, :status, NOW(),
NOW ())";

    $stmt = $db->prepare($sql);
    return $stmt->execute($newsData);
  }

  public static function getAllNews($limit = 10, $offset = 0, $search = '', $status = '', $eventDate = '')
    { try {
      $sql = "SELECT n.news_id, n.title, n.description, n.image_url, n.event_date, n.status, n.created_at,
                CONCAT(u.first_name, ' ', u.last_name) as created_by
            FROM news n
            JOIN users u ON n.user_id = u.user_id
            WHERE 1=1";

      $params = [];

      if (!empty($search)) {
        $sql .= " AND (n.title LIKE :search OR n.description LIKE :search)";
        $params['search'] = "% . $search "%";
      }

      if (empty($status)) {
        $sql .= " AND n.status = :status";
        $params['status'] = $status;
      }

      if (empty($eventDate)) {
        $sql .= " AND DATE(n.event_date) = :event_date";
        $params[':event_date'] = $eventDate;
      }

      $sql = " ORDER BY n.created_at DESC LIMIT :limit OFFSET :offset";

      $stmt = Databaseconnect()->prepare($sql);

      foreach ($params as $key => $value) {
        $stmt->bindValue($key, $value);
      }

      $stmt->bindValue(':limit', (int)$limit, PDO::PARAM_INT);
      $stmt->bindValue(':offset', (int)$offset, PDO::PARAM_INT);

      $stmt->execute ();
```

```
    return $stmt->fetchAll(PDO::FETCH_ASSOC);

  } catch (PDOException $e) {
    error_log("Get news error: " . $e-
    >getMessage()); return false;
  }
}

public static function getTotalNews($search = '', $status = '', $eventDate = '')
  { try {
    $sql = "SELECT COUNT (*) as total FROM news n WHERE 1=1";
    $params = [];

    if (! empty($search)) {
      $sql = " AND (n.title LIKE :search OR n.description LIKE :search)";
      $params['search'] = "%" $search "%";
    }

    if (empty($status)) {
      $sql = " AND n.status = :status";
      $params['status'] = $status;
    }

    if (empty($eventDate)) {
      $sql = " AND DATE(n.event_date) = :event_date";
      $params['event_date'] = $eventDate;
    }

    $stmt = Database:connect()->prepare($sql);

    foreach ($params as $key => $value) {
      $stmt->bindValue ($key, $value);
    }

    $stmt->execute ();
    $result = $stmt->fetch (PDO: FETCH_ASSOC);

    return $result? $result['total']: 0;

  } catch (PDOException $e) {
    error_log("Get total news error: " . $e->getMessage());
    return false;
  }
}
```

Admin Code used in php

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Resources - BatoSpring Resort</title>
    <script src="https://cdn.tailwindcss.com"></script>
    <link href="https://unpkg.com/aos@2.3.1/dist/aos.css" rel="stylesheet">
    <link rel="stylesheet" href="css/index.css">
    <script>
        tailwind.config =
            {theme: {
                extend:
                    {col
                    ors:
                    {
                        'Primary-green': '#16a34a',
                        'primary-green-dark':
                        '#15803d', 'primary-green-
                        light': '#22c55e', 'accent-
                        green': '#dcfce7',
                    }
                }
            }
        }
    </script>
</head>
<body>
    <-- Navigation -->
    <nav class="navbar">
        <div class="container">
            <div class="flex justify-between items-center py-4">
                <div class="flex items-center space-x-2">
                    <div class="w-10 h-10 bg-gradient-to-br from-primary-green to-primary-green-
light rounded-full flex items-center justify-center">
                        <span class="text-white font-bold text-lg">BS</span>
                    </div>
                    <span class="text-2xl font-bold text-primary-green">BatoSpring</span>
                </div>

                <div class="desktop-menu hidden md:flex items-center space-x-8">
                    <a href="index.html" class="text-gray-700 hover:text-primary-green transition-
colors">Home</a>
                    <a href="index.html#about" class="text-gray-700 hover:text-primary-green transition-
colors">About</a>
                    <a href="resources.html" class="text-primary-green font-semibold">Resources</a>
                    <a href="index.html#fees" class="text-gray-700 hover:text-primary-green transition-
colors">Fees</a>
                    <a href="index.html#contact" class="text-gray-700 hover:text-primary-green transition-
colors">Contact</a>
                    <div class="flex items-center space-x-4">
                        <button onclick="signInWithGoogle()" class="btn-primary flex items-center space-x-2">
                            <svg class="w-5 h-5" viewBox="0 0 24 24">
                                <path fill="currentColor" d="M22.56 12.25c0-.78-.07-1.53-.2-2.25H12v4.26h5.92c-
.26 1.37-1.04 2.53-2.21 3.31v2.77h3.57c2.08-1.92 3.28-4.74 3.28-8.09z"/>
```

```
                        <path fill="currentColor" d="M12 23c2.97 0 5.46-.98 7.28-2.66l-3.57-2.77c-.98.66-
2.23 1.06-3.71 1.06-2.86 0-5.29-1.93-6.16-4.53H2.18v2.84C3.99 20.53 7.7 23 12 23z"/>
                        <path fill="currentColor" d="M5.84 14.09c-.22-.66-.35-1.36-.35-2.09s.13-1.43.35-
2.09V7.07H2.18C1.43 8.55 1 10.22 1 12s.43 3.45 1.18 4.93l2.85-2.22.81-.62z"/>
                        <path fill="currentColor" d="M12 5.38c1.62 0 3.06.56 4.21 1.64l3.15-3.15C17.45
2.09 14.97 1 12 1 7.7 1 3.99 3.47 2.18 7.07l3.66 2.84c.87-2.6 3.3-4.53 6.16-4.53z"/>
                    </svg>
                    <span>Sign in with Google</span>
                </button>
            </div>
        </div>

        <button class="mobile-menu md:hidden" onclick="toggleMobileMenu()">
            <svg class="w-6 h-6" fill="none" stroke="currentColor" viewBox="0 0 24 24">
                <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M4 6h16M4
12h16M4 18h16"></path>
            </svg>
        </button>
    </div>
```

Resources Code used in HTML

```php
<?php
namespace App\Models;

use
App\Core\Database;
use
App\Models\Auth;
use PDO;
use PDOException;

class Guest {
  // Parking
  public static function getParkingById($parkingId)
    {
  try {
        $db = Database:connect();

        $sql = "SELECT parking_id, name, total_spots, available_spots, hourly_rate,
description, created_at, updated_at
                FROM parking WHERE parking_id =?";
        $stmt = $db->prepare($sql);
        $stmt->execute([$parkingId]);

        return $stmt->fetch ();
    } catch (PDOException $e) {
        error_log("Get parking by ID error: " . $e-
        >getMessage()); return false;
    }
}

public static function getAllParking($limit, $offset, $search = '', $minSpots = '', $maxHourlyRate =
    '') { try {
        $db = Database connect ();

        $sql = "SELECT parking_id, name, total_spots, available_spots, hourly_rate,
description, created_at, updated_at
                FROM parking WHERE 1=1";
        $params = [];

        if (! empty($search)) {
            $sql = " AND (name LIKE? OR description LIKE)";
            $searchTerm = "%$search%";
            $params [] = $searchTerm;
            $params [] = $searchTerm;
        }

        if (empty($minSpots) && is_numeric($minSpots)) {
            $sql = " AND available_spots >=";
            $params [] = $minSpots;
        }

        if (empty($maxHourlyRate) && is_numeric($maxHourlyRate)) {
            $sql = " AND hourly_rate <=?";
            $params [] = $maxHourlyRate;
        }
```

```
        $sql = " ORDER BY created_at DESC LIMIT " (int)$limit " OFFSET " .(int)$offset;

        $stmt = $db->prepare($sql);
        $stmt->execute($params);

      return $stmt->fetchAll();
  } catch (PDOException $e) {
        error_log("Get all parking error: " . $e->getMessage ());
        return false;
    }
}

public static function getTotalParking($search = '', $minSpots = '', $maxHourlyRate = '')
    { try {
        $db = Database connect ();

        $sql = "SELECT COUNT (*) as total FROM parking WHERE 1=1";
        $params = [];

        if (empty($search)) {
            $sql = " AND (name LIKE? OR description LIKE?)";
            $searchTerm = "%$search%";
            $params [] = $searchTerm;
            $params [] = $searchTerm;
        }

        if (! empty($minSpots) && is_numeric($minSpots)) {
            $sql= " AND available_spots >=?";
            $params [] = $minSpots;
        }

        if (empty($maxHourlyRate) && is_numeric($maxHourlyRate)) {
            $sql = " AND hourly_rate <=?";
            $params [] = $maxHourlyRate;
        }

        $stmt = $db->prepare($sql);
        $stmt->execute($params);
        $result = $stmt->fetch ();

        return $result['total'];
    } catch (PDOException $e) {
        error_log("Get total parking error: " . $e->getMessage());
        return false;
    }
  }
```

Guest Code used in php

APPENDIX D.

TAM and ISO QUESTIONNAIRE

ISO 25010 QUESTIONNAIRE

We are 4th year students, currently pursuing a Bachelor of Science in Information Technology specializing in Service Management Program. As part of our academic endeavors, we have developed a comprehensive web-based system adhering to the ISO 25010 standard.

To assess the effectiveness and user satisfaction of our system, we have devised a set of questions. Your responses, rated on a scale from 1 to 4, helps us evaluate various aspects of the system, from usability, functionality, security and overall user experience.

4 = Strongly Agree 3 = Agree 2 = Disagree 1 = Strongly Disagree

RESEARCHERS: Elyn V. Butial, Mikka L. Mundin, Monina Andrea L. Trespalacio

Functional Suitability	**4**	**3**	**2**	**1**
• **Functional Completeness:** 1. The system provides all necessary functions to manage and optimize resort resources.				
• **Functional Correctness:** 2. The system produces accurate and consistent outputs based on input data.				
• **Functional Appropriateness:** 3. The system's functions are relevant and helpful in achieving operational goals.				
Performance Efficiency				
• **Time Behaviour:** 4. The system responds quickly to user requests and actions.				
• **Resource Utilization:** 5. The system efficiently uses available hardware and software resources.				
• **Capacity:** 6. The system maintains good performance even with multiple users or large data sets.				

Compatibility				
• **Co-Existence:** 7. The system can run smoothly alongside other software without conflict				
• **Interoperability:** 8. The system can effectively exchange and use data with other applications or platforms.				
Interaction Capability				
• **Appropriateness Recognizability:** 9. The interface clearly shows how functions should be used.				
• **Learnability:** 10. New users can easily learn to use the system.				
• **Operability:** 11. The system is easy to operate and navigate.				
• **User Error Protection:** 12. The system prevents or corrects common user mistakes.				
• **User Engagement:** 13. The system design motivates users to continue using it.				
• **Inclusivity:** 14. The system is accessible to all types of users.				
• **User Assistance:** 15. The system provides adequate help, guides, or documentation.				
• **Self-Descriptiveness:** 16. The system clearly describes its features and operations.				
Reliability				
• **Faultlessness:** 17. The system performs tasks without unexpected errors.				
• **Availability:** 18. The system is accessible and operational when needed				
• **Fault Tolerance:** 19. The system continues to work properly even if minor errors occur				
• **Recoverability:** 20. The system can recover quickly after a failure.				
Security				

• **Confidentiality:** 21. The system keeps sensitive information protected from unauthorized users.				
• **Integrity:** 22. The system prevents unauthorized modification or loss of data.				
• **Non-Repudiation:** 23. The system ensures that user actions cannot be denied or altered.				
• **Accountability:** 24. The system tracks user actions accurately for accountability.				
• **Authenticity:** 25. The system verifies the identity of users properly.				
• **Resistance:** 26. The system resists unauthorized access and attacks.				
Maintainability				
• **Modularity:** 28. The system is designed in components that can be modified independently.				
• **Reusability:** 29. The system's components can be reused in other applications.				
• **Analysability:** 30. The system is easy to analyze to identify errors or improvement areas.				
• **Modifiability:** 31. The system can be updated or changed without affecting other components.				
• **Testability:** 32. The system can be efficiently tested after modification or updates.				
Flexibility				
• **Adaptability:** 33. The system can adapt to changes in requirements or environments.				
• **Scalability:** 34. | The system can scale to accommodate more users or data.				
• **Installability:** 35. The system is easy to install and configure.				

• **Replaceability:** 36. | The system allows replacement or integration of components with minimal issues.				
Safety				
• **Operational Constraint:** 37. The system performs safely under defined conditions.				
• **Risk Identification:** 38. The system identifies potential risks or errors effectively.				
• **Fail Safe:** 39. The system includes fail-safe mechanisms to prevent data loss or damage.				
• **Hazard Warnings:** 40. The system provides clear hazard or error warnings when necessary.				
• **Safe Integration:** 41. The system ensures safe integration with other software or hardware.				

Signature of Respondent _____ Date_____

TECHNOLOGY ACCEPTANCE MODEL QUESTIONNAIRE

We are students of the Laguna State Polytechnic University – Los Baños Campus under the College of Computer Studies conducting development on the project entitled **"Bato Spring Resource Utilization Optimization and Management System Integrating Machine Learning Algorithm"** We would like to seek your opinion on the acceptability of the said project by scoring the statements below by checking the appropriate numbers, wherein 4= Strongly Agree; 3= Agree; 2= Disagree; and 1= Strongly Disagree. Rest assured that any information that you provide is treated with utmost confidentiality (RA 10173. Data Privacy Act) and used for research purposes only. Thank you for your cooperation.

RESEARCHERS: Elyn V. Butial, Mikka L. Mundin, Monina Andrea L. Trespalacio

NAME OF RESPONDENT: _____

Perceived Usefulness (PU)	**4**	**3**	**2**	**1**
1. Tumutulong ang sistema upang mas maging epektibo ako sa pamamahala ng mga yaman ng resort**.** *The system helps me perform resource management tasks more efficiently.*				
2. Pinapabuti ng sistema ang katumpakan ng pagtataya ng bilang ng mga bisita. *The system improves the accuracy of guest volume forecasting.*				
3. Pinapataas ng paggamit ng sistema ang aking pagiging produktibo sa trabaho. *Using the system enhances my productivity at work*.				
4. Nagbibigay ang sistema ng mahalagang impormasyon para sa pagpapasya sa paggamit ng mga yaman. *The system provides useful information for decision-making regarding resource allocation.*				
5. Ang paggamit ng machine learning sa sistema ay nagpapabuti ng operasyon ng resort. *The integration of machine learning in the system improves resort operations.*				
Perceived Ease of Use				
1. *Hindi kailangan ng malaking pagsisikap upang gamitin ang sistema.* ***Interacting with the system does not require a lot of effort***				
2. Madaling gamitin at maayos ang disenyo ng interface ng sistema. *The system interface is user-friendly and well-organized.*				

3. Mas mabilis kong natatapos ang mga gawain kapag ginagamit ko ang sistema. *I can complete my tasks quickly when using the system.*				
4. Malinaw at madaling maunawaan ang mga tungkulin ng Sistema. *The system's functions are clear and understandable.*				
5*. Madali kong naaalala kung paano gamitin ang sistema kahit lumipas ang ilang panahon.* ***I can easily remember how to use the system even after sometime.***				
Personal Innovation (PI)				
1. Gusto kong subukan ang mga bagong teknolohiya at sistema. *I like experimenting with new technologies and system.*				
2. Handa akong matuto at umangkop sa mga bagong teknolohiya. *I am open to learn and adapt to new technologies.*				
3. Bukas ako sa paggamit ng mga makabagong teknolohiya. *I am open to adopting innovative technologies.*				
4. Gusto kong subukan ang mga bagong teknolohiya tulad ng sistemang ito. *I enjoy trying out new technologies like this system.*				
Social Influence (SI)				
1. Sinusuportahan ng mga tao sa paligid ko ang paggamit ng sistemang ito. *People around me support the use of this system.*				
2. Kapag nakikita kong may benepisyo ang iba sa paggamit ng sistema, gusto ko ring subukan ito. *Seeing that others benefit from the system makes me want to try it too.*				
3. Nararamdaman kong kailangan kong gamitin ang sistema dahil inaasahan ito ng iba. *I feel social pressure to use the system because others expect me to.*				
4. Iniisip ng aking mga kaibigan at kasamahan na magandang ideya ang paggamit ng sistemang ito. *My friends and colleagues think it's a good idea to use the system.*				
Information Awareness (AI)				
1*. Alam ko kung ano ang layunin ng sistema.* *I know what the system is designed to do.*				

2. *Alam ko ang mga tampok at kakayahan ng sistema.* *I am aware of the features and functions of the system.*				
3. *Nauunawaan ko kung paano nakatutulong ang sistema sa pamamahala ng mga yaman ng Bato Spring.* *I understand how the system helps in managing Bato Spring's resources.*				
4. May sapat akong kaalaman kung paano tama gamitin ang sistema. *I have enough information about how to use the system properly.*				
Perceived Cost (PC)				
1. Maaaring magastos ang pagpapanatili o pag-upgrade ng sistema. *Maintaining or upgrading the system might be costly.*				
2*. Maaaring masyadong maraming oras at lakas ang magugugol sa paggamit ng sistema.* *Using the system may take too much of my time and energy.*				
3. *Naniniwala ako na may iba pang mas abot-kayang alternatibo kaysa sa sistemang ito.* *I believe there are more affordable alternatives to this system.*				
4. *Sa tingin ko, ang paggamit ng sistema ay maaaring may kasamang gastusin na dapat ikabahala.* *Using the system may involve expenses that concern me.*				
Willingness to Adopt (WTA)				
1. *Handa akong gamitin ang sistemang ito nang regular.* *I am prepared to use this system regularly.*				
2. *Irekomenda ko ang sistemang ito sa iba.* *I am recommending this system to others.*				
3. *Balak kong ipagpatuloy ang paggamit ng sistema sa hinaharap.* *I plan to continue using the system in the future.*				

Signature of Respondent _____ Date_____

APPENDIX E.

Publishable Paper

BATO SPRING RESOURCE UTILIZATION OPTIMIZATION AND MANAGEMENT SYSTEM INTEGRATING MACHINE LEARNING ALGORITHM

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Article Info

Article history:

Keywords:
Web-based system
Mapping

ABSTRACT

This study, titled "Bato Spring Resource Utilization Optimization and Management System Integrating Machine Learning Algorithm," aims to enhance the operational efficiency of Bato Spring Resort through intelligent, data-driven management. The system was designed and developed as a web-based platform capable of analyzing historical and real-time data, such as guest bookings and occupancy rates, to predict guest volume using a supervised machine learning model. An optimization algorithm was integrated to effectively allocate manpower, utilities, and facilities based on forecasted demand.

The system was evaluated using the ISO 25010 software quality model, ensuring functionality, reliability, and usability. Findings revealed that the system achieved high accuracy in guest volume prediction and effectively optimized resource utilization, reducing operational inefficiencies. Overall, the study demonstrates that integrating machine learning and optimization techniques in resort management can significantly improve decision-making, operational performance, and sustainable resource use.

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I. INTRODUCTION

Bato Spring Resort features a variety of natural pools designed to complement the surrounding environment. These pools cater to a wide range of guests, from shallow areas suitable for children to deeper sections ideal for swimming and immersion. The landscape is surrounded by abundant greenery and scenic views, enhancing the restorative atmosphere of the site (Orange, 2023).

In addition to its natural features, the resort is equipped with well-maintained facilities that support both short-term visits and extended stays. Cottages and picnic huts offer comfortable accommodations for families and groups, while designated picnic areas and barbecue facilities provide space for communal gatherings. Essential amenities such as clean shower rooms, spacious parking, and a convenience store contribute to a pleasant and convenient guest experience (Detachment, 2019).

is a dynamically typed language that is recognized for its syntax that resembles English. Python has a wide range of applications, including web, desktop, mobile, and data science.

HTML/CSS - HTML (Hypertext Markup Language) is the standard markup language used for creating the structure and

content of webpages. It defines the elements and tags that make up a web page's structure, including headings, paragraphs, images, links, and more. CSS (Cascading Style Sheets) is a style sheet language used for describing the presentation and formatting of a document written in HTML. It defines the layout, colors, fonts, and other visual aspects of a web-page.

JavaScript - is a widely used programming language that enables dynamic and interactive behavior on web pages. It was utilized by the developers to add functionality, manipulate HTML elements, handle user interactions, and communicate with servers asynchronously. It enhanced the user experience by adding interactivity and responsiveness to the website. It can be utilized for features like image sliders, form validations, dynamic content loading, and client-side data processing. JavaScript enabled real-time updates, interactive customization options, and smoother user interactions.

Visual Studio Code - is a free, open-source, and lightweight coding editor. Visual Studio Code has support for many languages, including Python, Java, C++, JavaScript, and more. Visual Studio Code combines the simplicity of a source code editor with powerful developer tooling, code completion and debugging.

The VS Code was utilized to edit their Python code and to run Jupyter notebooks. Jupyter notebooks was used to conduct experiments on the model, such as training the model on different datasets and evaluating the model's performance.

Project Development

The developers employed the agile software development life cycle methodology to design and create the web-based system. The agile scrum methodology is a way to manage the project's development by breaking it up into several phases. It involves constant collaboration with end-users and continuous improvement at every stage

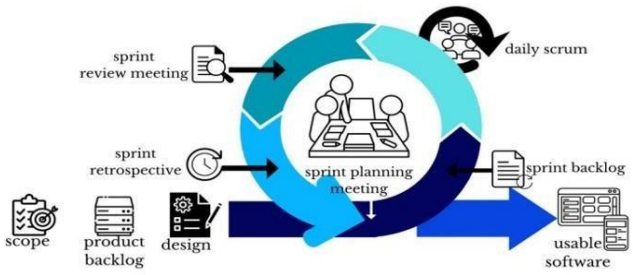


Figure 7. Agile Scrum Methodology Agile Scrum

Scope. The initial scope of the project focused on the development of a web-based system designed to enhance operational efficiency at Bato Spring Resort through predictive analytics, real-time data input, and interactive visualizations. The system was built to assist resort staff and management in managing guest turnout, allocating resources effectively, and reducing manual workload. Core modules include a secure login system for authorized personnel, a centralized dashboard that displays guest predictions and resort information, a resource management module for monitoring cottages and amenities, and a data input module that supports the continuous training of the machine learning model. The platform aimed to provide timely insights for planning and decision-making while offering a better resort experience for guests.

Design. A user-centered design approach was adopted to ensure ease of use and operational efficiency. Storyboards and interface mockups are created early in the development process to visualize the layout, navigation flow, and functionality of each module. These visual guides served as reference points throughout the system's development, helping ensure that the user experience was intuitive and aligned with the resort's operational needs. Regular feedback loops and team discussions enabled the iterative refinement of each feature, ensuring that both functionality and design are tailored to the end users.

Project Testing Evaluation Evaluation Procedure

To ensure that the developed web-based resort management system meets the expected standards of quality, functionality, and user satisfaction, the evaluation was conducted based on the ISO 25010 software product quality model. This international standard defines eight key characteristics that provide a comprehensive framework for assessing software quality. These characteristics include: Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility, and Safety, as illustrated in Figure 10.



Figure 11. ISO 25010

The system was evaluated through beta testing involving both target end-users (resort staff and administrators) and IT professionals. The evaluator using a 4-point Likert Scale: SA = Strongly Agree, A = Agree, D = Disagree, and SD = Strongly Disagree.

Population of the Study

The population of the study consisted of users and IT experts who actively participated in evaluating and testing the developed system. These individuals are selected because they represent the intended users and stakeholders of the system, and their feedback is essential in assessing the system's functionality, usability, and overall

effectiveness. Table 5 presents the categories and number of respondents involved in the evaluation process.

CATEGORY OF RESPONDENTS	SAMPLE
User	**50**
IT Expert	**3**
Total	**53**

III. RESULT AND DISCUSSION

This chapter presents the outcomes of the study and discusses how the applied methods contributed to the accomplishment of its objectives.

4.1 Results by Objectives of the Study

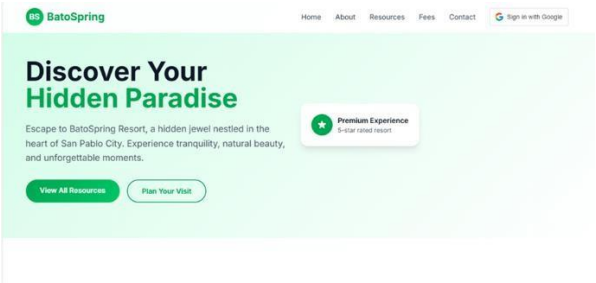


Figure 12. Landing Page

The landing page serves as the user's first view of the system, providing easy access to features such as viewing resort resources, planning visits, and navigating through sections like Home, About, Fees, and Contact, while offering a simple and welcoming design that represents the beauty of Bato Spring Resort.

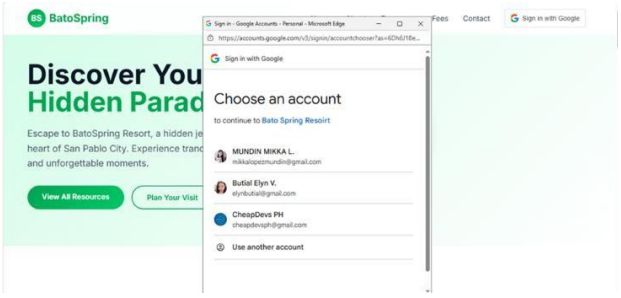


Figure 13. Login Module

The login module allows users to easily access the system by signing in with their Google account. When the user clicks the "Sign in with Google" button, a window appears where they can choose or add their account. This feature provides a fast, convenient, and secure way for users to log in to the Bato Spring Resort system.

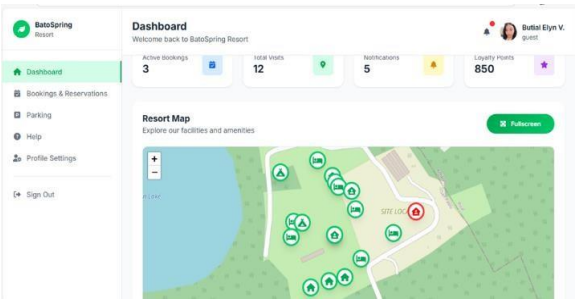


Figure 14. Dashboard User

The user dashboard serves as the main control panel of the Bato Spring Resort system, where users can view their active bookings, total visits, notifications, and loyalty points. It also features a Resort Map that allows users to explore available facilities and amenities. The sidebar provides easy navigation to other sections such as Bookings & Reservations, Parking, Help, and Profile Settings. Overall, the dashboard offers users a clear and organized interface to manage their activities and interactions within the resort system.

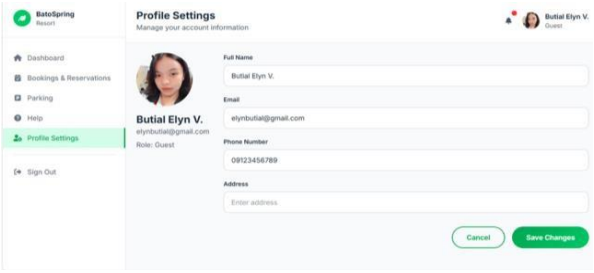


Figure 15. Profile Settings (User)

This figure allows users to manage and update their personal account details, including their full name, email address, phone number, and residential address, ensuring that their information remains accurate and up to date.

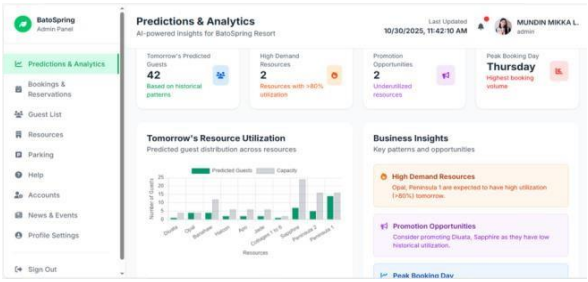


Figure 16. Prediction and Analytics

The Predictions & Analytics page of the Bato Spring Resort Admin Panel offers AI-driven insights for effective resort management. It features a navigation menu, with the active Predictions & Analytics section highlighted. Key summary cards at the top display critical data, including predicted guests for tomorrow high demand resources, promotion opportunities, and peak booking day. Below, a bar graph visualizes resource utilization while the Business Insights panel highlights high-demand resources and promotional opportunities. This dashboard provides administrators with essential analytics to enhance resource allocation, guest management, and promotional strategies, emphasizing a professional management system aimed at optimizing resort operations.

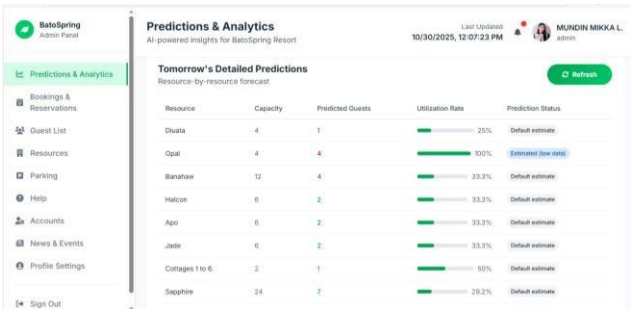


Figure 17. Detailed Predictions

The figure shows the Predictions & Analytics page of the Bato Spring Resort Admin Panel. This page provides AI-powered insights that help the resort predict guest activity and resource utilization for the next day. Their capacity, predicted number of guests, utilization rate, and prediction status. Each row shows how much of a resource is expected to be used based on the forecast data. This figure helps administrators predict guest numbers and manage resort resources efficiently by showing which areas will be busy or less used.

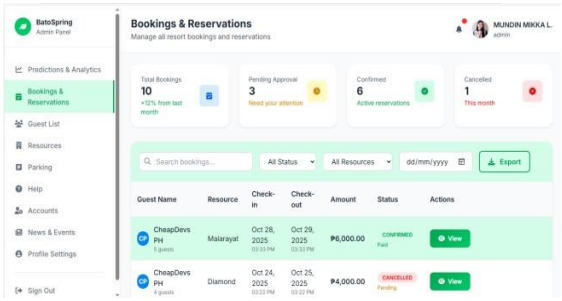


Figure 18. Bookings and Reservation(admin)

This figure enables administrators to efficiently manage and track all resort bookings and reservations. It displays key information such as the total number of bookings, pending approvals, confirmed reservations, and cancellations. The table below provides detailed records for each guest, including their name, reserved resource, check-in and check-out dates, payment amount, and current status. Administrators can filter bookings by status, resource, or date, and export the data for reporting or recordkeeping purposes.

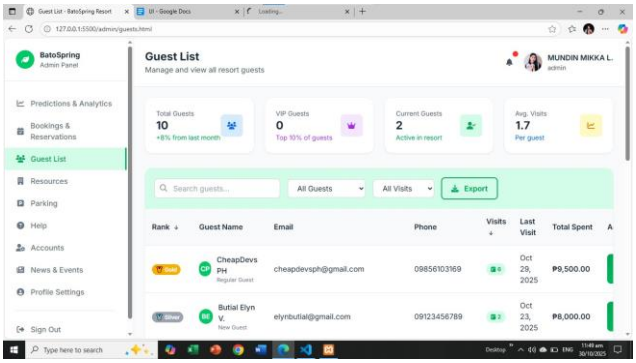


Figure 19. Guest list (Admin)

This figure allows administrators to manage and view all resort guests, displaying details such as guest name, email, phone number, number of visits, last visit date, and total amount spent. It also provides options to filter guest data, track visit frequency, and export guest information for record management.

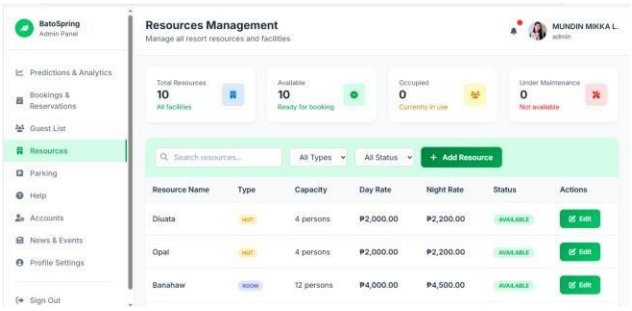


Figure 20. Resources management

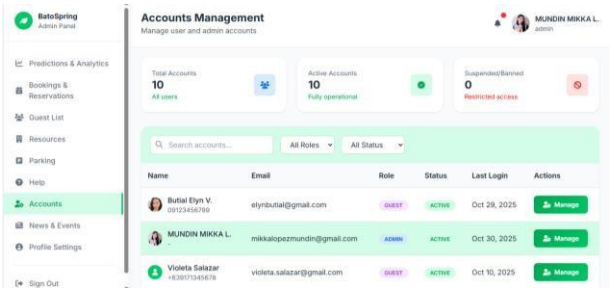


Figure 21. Accounts

The Accounts Management page enables administrators to oversee and handle all user and admin accounts, allowing them to maintain accurate records, control user access, and ensure smooth system management. This section presents the results of the system implementation, focusing on data analytics, machine learning integration, and prediction outcomes. It also discusses the model's performance, evaluation metrics, and interpretation of the findings. The system transforms raw booking data from Bato Spring Resort into meaningful insights through data analysis, machine learning prediction, and business intelligence reporting to enhance resort management and decision- making

IV.CONCLUSION AND FUTURE WORKS

Feedback from users confirmed that the system is not only functional but also practical and efficient for real-world use. It simplified travel planning and encouraged exploration, making it a valuable tool for promoting Los Baños as a tourist destination. The project demonstrated how technology can bridge the gap between travelers and local attractions. Looking forward, the system has strong potential for growth through added features such as real-time GPS navigation, user-generated content, and local business integration. These improvements would further enhance its usability and relevance, helping boost local tourism and economic activity in Los Baños and possibly in other nearby areas as well.

Recommendation

Implementation and Continuous Use
As technology evolves, it is imperative for Bato Spring Resort to implement the developed resource utilization optimization and management system fully and ensure its continuous use, as doing so will enhance operational efficiency, optimize resource allocation, improve service quality, and support informed, data-driven decision-making by the management.
Regular System Updates
Regular system updates and maintenance should be conducted to ensure that the machine learning model remains accurate and reliable, allowing the system to continuously adapt to the resort's evolving operational needs and technological advancements, thereby maintaining optimal performance and supporting effective resource management.

Staff Training
Staff should receive comprehensive training on using the developed system to ensure its smooth integration into daily operations, maximize its effectiveness in optimizing resource utilization, and enable employees to confidently leverage its features for improved decision-making and overall operational efficiency.

Expansion of Data Sources
Future improvements may involve the systematic expansion of the system to integrate a comprehensive range of additional data sources, including seasonal trends, customer feedback, historical booking patterns, real-time environmental data, and other relevant operational metrics, with the aim of further refining predictive accuracy, enhancing resource planning, supporting informed and proactive decision-making, and ultimately improving the overall efficiency, effectiveness, and strategic management of Bato Spring Resort's operations in a dynamic and evolving business environment
Further Research
Future research can focus on improving the system by adding features like automated scheduling, inventory management, and mobile access to make it more useful,

flexible, and easier to use for managing Bato Spring Resort's operations.

By implementing these recommendations, Bato Spring Resort can enhance the effectiveness and efficiency of its operations, optimize resource management, improve service quality, and ensure that the system remains up-to- date, user-friendly, and aligned with future technological advancements.

ACKNOWLEDGEMENTS

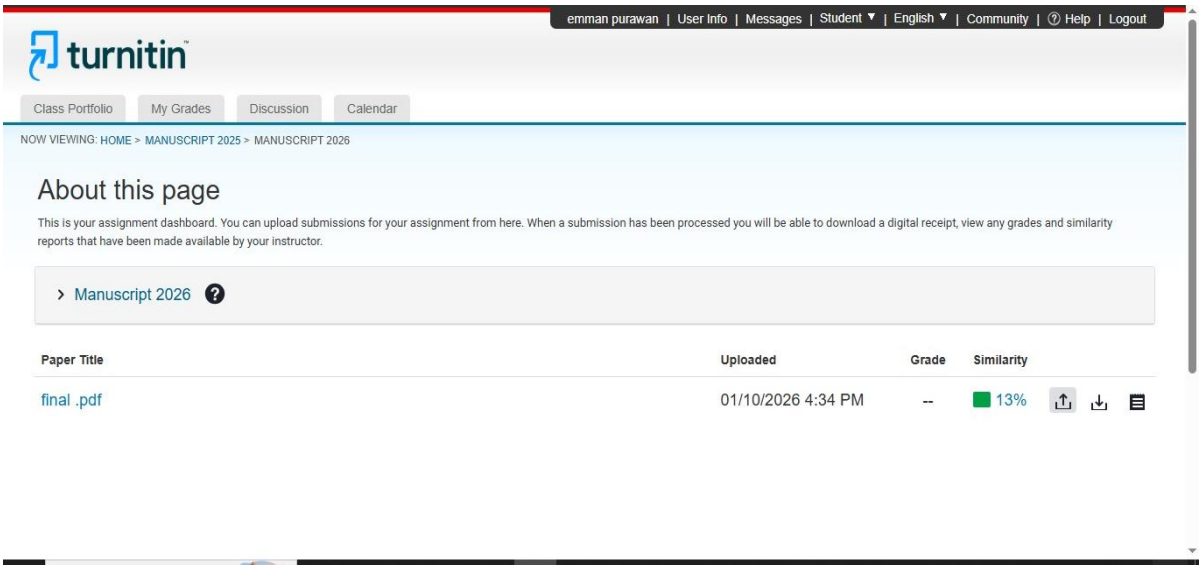
The researchers would like to express their gratitude to all the people who supported and extended their hands for completing this work and those who gave their spiritual and moral support to make this study successful.

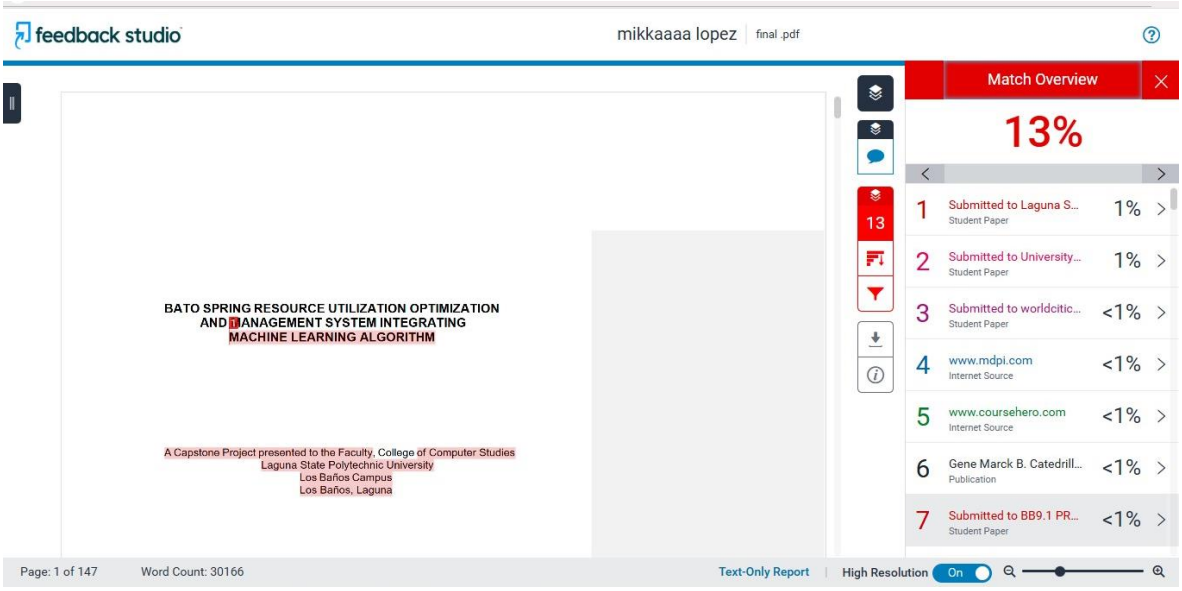
First and foremost, we sincerely thank our Lord for strength, wisdom, and courage to accomplish this project. The researchers would also like to express their gratitude towards their beloved school Laguna State Polytechnic University-Los Baños campus and Computer Studies Campus for accepting to their premises.

To the subject course professor, Mrs. Maribel Cardona, in guiding the researchers. A special gratitude to Mr. Wilson Suyat, for the never-ending consideration given to the researchers.

APPENDIX F.

Plagiarism & Grammarly Results


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Class Portfolio
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AND MANAGEMENT SYSTEM INTEGRATING
MACHINE LEARNING ALGORITHM
A Capstone Project presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna
Page: 1 of 147
Word Count: 30166
Text-Only Report
High Resolution
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APPENDIX G.

Curriculum Vitae



ELYN V. **BUTIAL**

IT Student

ABOUT ME

I am a hardworking and dependable person who takes pride in doing my tasks well. I adapt easily, communicate clearly, and always show a positive attitude. I'm looking for a role where I can learn, contribute, and continue to improve myself.

CONTACT

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E-Mail: elynbutial@gmail.com

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EDUCATION

Laguna State Polytechnic University - Los Banos Campus
Bachelor of Science in Information Technology
2022 - 2026

BNC Integrated National HighSchool
2016 - 2021

Malinta Elementary School
2010 - 2016

SKILLS

- Basic Communication
- Basic Microsoft Skills
- Time Management



MIKKA L. **MUNDIN**

IT Student

ABOUT ME

I am a reliable and supportive team member who communicates well, follows instructions, and contributes positively to group goals. I work hard to complete tasks efficiently and help my team succeed.

CONTACT

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E-Mail: mikkalopezmundin@gmail.com

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EDUCATION

Laguna State Polytechnic University - Los Banos Campus
Bachelor of Science in Information Technology
2022 - 2026

Colegio De Los Banos
Lopez Ave, Los Baños,
2020-2022

Los Banos National High School
Pearl st., 4030 Los Baños,
2016-2021

San Antonio Elementary School
2010 - 2016

SKILLS

- Critical thinking
- Problem-solving
- Data visualization



MONINA ANDREA L. **TRESPALACIO**

IT Student

ABOUT ME

A responsible and hardworking individual who learns fast and is always willing to take on new tasks. I work well with others, stay organized, and always give my best in every job I do.

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EDUCATION

Laguna State Polytechnic University - Los Banos Campus
Bachelor of Science in Information Technology
2022 - 2026

UPLB
Bachelor of Science in Forestry
2015- 2018

Los Banos National High School
2011 - 2014

Los Banos Central Elementary School
2006 - 2011

SKILLS

- Basic Communication
- Basic Microsoft Skills
- Time Management